



# R&D dynamics and corporate cash saving

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## ABSTRACT

High-tech firms hold substantial cash reserves. We build a parsimonious industry equilibrium model with endogenous productivity to study high-tech firms' cash-saving policy and explore its role in innovation. The model incorporates multiple well-cited explanations in the literature and examines which factors are the main determinants of high-tech firms' saving behavior. We find that innovation uncertainty is the major driver, followed by knowledge spillover and financial frictions. Market competition has non-monotonic effects, while R&D adjustment costs play a relatively minor role. We also find that even if the productivity process is transitory, firms manage to save enough to reduce R&D distortions from financial frictions.

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## 1. Introduction

Non-financial businesses in the United States hold 38% of monetary aggregate M1 and 22% of monetary aggregate M2. Among them, publicly traded high-tech firms maintain a sizable level of cash reserves (Bates et al., 2009; Begenau and Palazzo, 2021). Their median and mean cash-to-assets ratio are 0.32 and 0.37, respectively, during the 2000s. In light of the fact that high-tech firms are key drivers of innovation, productivity, and economic growth, it is important to understand the determinants of their saving behavior and then to explore the role of corporate saving in innovation.

Previous studies attribute the large cash stocks to various factors relevant for R&D intensive industries: R&D adjustment costs (Brown and Petersen, 2011), financial frictions caused by the non-collateralizability of R&D (Falato et al., 2013), knowledge spillover (Qiu and Wan, 2015), innovation uncertainty (Lyandres and Palazzo, 2014; Hsu et al., 2016), and intense market competition (Ma et al., 2020; Lyandres and Palazzo, 2016). One of the primary goals of our paper is to consider all

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of these possible explanations in the literature, identify the main determinant(s) of high-tech firms' strong cash demand, and quantify their impacts.<sup>2</sup>

In addition, the role of corporate saving in mitigating the distortion caused by financial frictions remains inconclusive and depends on the persistence of productivity processes (Buera et al., 2011; Midrigan and Xu, 2014; Moll, 2014). In particular, the distortion-correction role of corporate saving is effective only if productivity is sufficiently persistent. This paper re-examines this issue in the context of high-tech industries in which productivity is endogenized.

Toward this end, we build a dynamic industry equilibrium model of R&D and cash with endogenous entry and exit. In the model, firms hire labor to produce differentiated products and compete in the product market. They invest in R&D to improve their competitive positions, but face knowledge spillover and innovation risks. Also, adjusting R&D expenditures incur costs. The capital market is assumed to be imperfect, so firms have no access to debt financing due to prohibitive collateralization of R&D; they can, however, raise funds through equity issuance (Brown et al., 2009; Hall and Lerner, 2010; Hall et al., 2016). Equity inflows (equity issuance) and equity outflows (dividend distribution) are costly. This generates a precautionary motive for holding cash, as in Zhao (2020). That is, to avoid external financing costs in case of a liquidity shortage, firms accumulate internal funds. Meanwhile, the presence of other frictions—R&D adjustment costs, knowledge spillover, innovation uncertainty, and market competition—affects the strength of the precautionary motive through the cost of cash saving and the amount of cash flows. But to be clear, the precautionary motive we have in mind here is quite broad. Internal funds can also be used to respond more efficiently to positive shocks.

To examine the model's quantitative implications for research intensive firms' cash-saving policies, we estimate the model and validate it by demonstrating its ability to match relevant data moments. We then use the structural estimates obtained to assess how each factor affects cash and R&D choices jointly. Each factor is encapsulated by one or more estimated parameters. We change one parameter at a time and perform several sets of counterfactual analyses: (i) increasing the intensity of each mechanism by 5%, (ii) increasing the intensity of each mechanism such that mean firm value changes by 1%, and (iii) shutting down each channel in turn. Interestingly, shutting down any one of the main channels can produce considerable effects on cash—this is probably why there are such diverse explanations in the literature. However, at the margin, the relative strengths of the mechanisms are quite different.

Innovation uncertainty has the strongest effect, and knowledge spillover and financial frictions are also significant drivers behind high-tech firms' strong cash demand. These results support the empirical findings reported by Falato et al. (2013), Qiu and Wan (2015), and Hsu et al. (2016). However, our results deviate from previous studies along several dimensions. First, the endogenized productivity process allows us to decompose and interpret innovation uncertainty at a deeper level. Our model differentiates between innovation volatility and efficiency which are both under the umbrella of innovation uncertainty, and we show that the spread of the conditional productivity shock is the key to explain the large cash balance. In addition, knowledge spillover affects cash policy through a channel different from that hypothesized by Qiu and Wan (2015). In particular, our model suggests that in response to an increase in knowledge spillover, firms cut R&D investments. The decline in precautionary cash demand, however, is fully offset by the increase in the substitution away from R&D to cash (substitution effect) and the increase in income that spillover brings (income effect). This generates a pattern in line with the somewhat surprising empirical observation that cash holdings increase with knowledge spillover.

R&D adjustment costs play a smaller role in generating cash holdings than the other factors, because the absence of adjustment costs only reduces the real price of investment—and thus affects the level of R&D expenditures. As long as financial frictions and income uncertainty are present, cash will be valuable regardless of the magnitude of the adjustment costs. R&D investment is also naturally persistent since productivity improvements made by R&D are stochastic. Firms are never certain their R&D investment will successfully lead to higher productivity, so they also have no reason to change the R&D level unless there are big enough shocks.

Market competition has important but non-monotonic impacts on high-tech firms' cash policy, regardless of the sources of competitive pressure—higher product substitutability or fixed operating costs. The non-monotonicity arises from the interaction of different forces. There is higher cash demand due to the precautionary motive when market competition is greater. But firms also tend to increase R&D spending as market competition intensifies. As a result, cash savings drop and exits rise precipitously when market competition becomes too large since firms still want to maintain their level of R&D expenditures. More precisely, the substitution from cash to R&D and the income effect from lower cash flows overpower the precautionary motive to save.

We further examine the role of corporate saving in influencing high-tech firms' R&D investment. Our result partially deviates from that derived from previous macroeconomic studies. We find that corporate saving corrects the distortion caused by financial frictions, and the efficacy of self-financing does not depend heavily on productivity persistence. This key difference is generated from our model assumption which reflects the empirical facts that large firms are the main providers of business R&D in the United States and they can issue equity to save (Eisfeldt and Miur, 2016). This assumption disconnects savings from income and allows firms to accumulate cash even if the productivity process is not sufficiently persistent.

<sup>2</sup> In addition to the factors mentioned above, Foley et al. (2007) also link high cash holdings with tax policy. They find that multinationals facing higher repatriation taxes tend to store more cash. We compare the average cash ratios for high-tech firms with and without foreign income. Interestingly, multinational high-tech firms tend to have lower cash-to-assets and cash-to-sales ratios than their counterparts: 0.33 versus 0.40 and 0.86 versus 2.51, respectively. It seems reasonable to conjecture that factors other than the tax-avoidance motive are responsible for high-tech firms' cash stock. As such, we abstract from a tax-based explanation in this paper.

Investors are also willing to support low-productivity but research-intensive firms. Over 50% of firms have negative net income in our sample of high-tech firms. Similarly, in our model with endogenous productivity, it is rational to keep high R&D firms with low productivity operating in most cases.

This paper contributes in several ways. First, it complements corporate cash studies by systematically analyzing the determinants of high-tech firms' substantial cash reserves. Previous structural work mainly focuses on average firms and examines their saving motives (Armenter and Hnatkovska, 2016; Boileau and Moyen, 2016; Nikolov and Whited, 2014; Riddick and Whited, 2009; Begenau and Palazzo, 2021). For high-tech firms, a number of factors have been empirically identified; however, little is known about their quantitative importance, which is the key step toward understanding high-tech firms' cash-saving decisions as well as the role of corporate saving in innovation. Furthermore, these factors have been examined in isolation, and this may cause erroneous conclusions to be reached. We show in our study that some results from previous work are overturned once all the features are analyzed jointly.

Second, our paper explores the role of firms' self-financing in mitigating distortions from financial frictions when productivity dynamics are endogenized, while most previous work assumes a stochastic productivity process which is beyond the control of firms (Buera et al., 2011; Midrigan and Xu, 2014; Moll, 2014). Our paper has the potential to deliver a deeper understanding of the importance of corporate savings in correcting misallocation. Instead of analyzing the issue through the traditional channel—physical capital allocation across firms and firms' entry and exit decisions, our paper emphasizes the channel through firms' R&D investment which has direct impacts on productivity dynamics.

Third, our paper develops a novel and computationally tractable framework to study firms' R&D choices. Specifically, it proposes a theoretically appealing approach to model the transition probabilities that govern the evolution of endogenous productivity. In contrast to previous studies that restrict firms from moving beyond the technology level that is closest to their current one (Ericson and Pakes, 1995; Xu, 2008; Hashmi and Van Biesebroeck, 2016), our approach allows for the realization of different levels of innovation—radical innovation vs. incremental innovation—and neatly captures a wide range of possible innovation dynamics. In the process of the estimation of our model, we also provide some of the first structural estimates for R&D adjustment costs.

The remainder of the paper is structured as follows. Section 2 lays out a dynamic industry equilibrium model of R&D and cash with endogenous entry and exit. Section 3 provides intuition about how considered R&D features influence high-tech firms' cash and R&D choices and reports the estimation results of the model. Section 4 quantitatively evaluates the importance of each factor in explaining high-tech firms' cash demand and discusses the role of corporate savings in high-tech industries. Section 5 concludes.

## 2. Model

In this section, we consider an industry equilibrium model of corporate R&D investment and cash savings, which nests Hopenhayn (1992), Riddick and Whited (2009), and Xu (2008). In the model, time is discrete and infinite. Within the industry, a continuum of firms with mass one operate in a monopolistically competitive market and specialize in the production of differentiated goods. Firms compete in the product market and face financial frictions. Each period, incumbent firms make their exit decisions.<sup>3</sup> Firms that choose to continue to operate produce goods and make financial, R&D-investment, and dividend-distribution choices. Potential entrants make their entry decisions by weighing the expected benefits and costs, and firms that eventually choose to enter decide their R&D spending.

We first specify the demand function that each firm faces and its production technology, knowledge accumulation process, and financing options, then state the firm's problem and the industry equilibrium.

### 2.1. Demand and technology

Assume there is a unit measure of firms and firm  $i \in [0, 1]$  produces output with labor  $l_i$ ,

$$y_{i,t} = z_{i,t} l_{i,t}. \quad (1)$$

Here,  $z_{i,t}$  is the firm's relative technological position within the industry at period  $t$  and is determined stochastically and is dependent on the firm's investment in R&D, its past technological position, and the intra-industry knowledge spillover.

We do not include physical capital in our model. This simplifying assumption is unlikely to affect our model's key implications for a variety of reasons. First, the capital-to-assets ratio of R&D intensive firms is only about one third as much and exhibits less autocorrelation when compared to R&D non-intensive firms. Next, R&D intensity is almost independent of physical capital stock for high-tech firms.<sup>4</sup> Cash flow and equity issuance are the two main sources of funds for high-tech firms' internal cash, and their dynamics are mainly driven by R&D investment rather than capital investment. For example, the correlation between R&D intensity and the future cash flow-to-assets ratio is ten times larger than the correlation

<sup>3</sup> We model firm entry and exit endogenously, because an exogenous entry and exit assumption can force high-productivity firms to exit, which is not a desirable feature. Firms also need the exit option because cash flow and hence firm value can become negative.

<sup>4</sup> The correlation between the R&D-to-sales ratio and beginning-of-period net capital stock is -0.067 for high-tech firms during the 2000s. See, also, Klette and Kortum (2004).

between the capital investment rate and future cash flow. Also, R&D intensity is strongly correlated with the current-period cash flow ratio, while the capital investment rate is almost independent of the latter. Finally, the static productive response to endogenous productivity changes in the model mimics the quick and efficient outsourcing of production by innovative firms in the real world. We do recognize, however, that introducing physical capital in our model as well as any other feature could change the dynamics and results.

The firm faces a downward sloping demand function according to the price that it sets  $p_{i,t}$  and this function takes the following form:

$$y_{i,t} = \left( \frac{p_{i,t}}{P_t} \right)^{-\sigma} Y_t, \quad (2)$$

where  $P_t = \left[ \int_0^1 p_{i,t}^{1-\sigma} di \right]^{\frac{1}{1-\sigma}}$  and  $Y_t = \left[ \int_0^1 y_{i,t}^{\frac{\sigma-1}{\sigma}} di \right]^{\frac{\sigma}{\sigma-1}}$ . Here,  $\sigma$  denotes the elasticity of substitution between goods,  $p_{i,t}$  denotes the price charged by firm  $i$ ,  $P_t$  and  $Y_t$  are industry price and quantity levels, and  $y_{i,t}$  is the quantity demanded for good  $i$ . This demand curve can be derived from the optimal choices of households.<sup>5</sup> They consume a composite consumption good that is a Dixit-Stiglitz index of differentiated goods in the industry.

## 2.2. Knowledge investment and production

In every period  $t$ , firm  $i$  invests in R&D and learns via spillover from competing firms that have relatively more advanced technologies. The firm's total input of knowledge production,  $D_{i,t}$ , at period  $t$  is therefore given by

$$D_{i,t} = d_{i,t} + \theta S_{i,t-1}. \quad (3)$$

Here,  $d_{i,t}$  is firm  $i$ 's own R&D spending,  $\theta$  captures the magnitude of knowledge spillover, and  $S_{i,t-1}$  is the average previous-period knowledge investment by firms that have stronger technological positions than firm  $i$  in the current period, that is,  $S_{i,t-1} = \int_{\{j: z_{j,t} > z_{i,t}\}} d_{j,t-1} dj$ . From this point on, we drop subscript  $i$  to simplify notation.

The firm must pay costs for adjusting its knowledge investment level, defined as

$$A(d_{t-1}, d_t) = \gamma_0 \mathbf{1}_{d_t \neq d_{t-1}} + \gamma_2 \frac{(d_t - d_{t-1})^2}{d_t + d_{t-1}}, \quad (4)$$

where  $\gamma_0 \geq 0$  is the fixed adjustment cost,  $\gamma_2 \geq 0$  is the convex adjustment cost, and  $\mathbf{1}_{d_t \neq d_{t-1}}$  is an indicator function that equals one if current R&D investment deviates from previous R&D spending. Adjustment costs can be justified by the fact that a large share of R&D expenses goes to scientists' salaries. Building a large team of skilled workers is costly since it often takes a while to get started on or acquainted with complicated projects. On the other hand, the turnover of scientists can also be very disruptive (Hall, 1993; Bond et al., 2005). This specification is close to the one assumed for physical capital adjustment costs (Abel and Eberly, 2002; Christiano et al., 2005; Schmitt-Grohe and Uribe, 2005; Cooper and Haltiwanger, 2006; and Del Negro and Schorfheide, 2008, among others).

The fixed cost reflects the loss from disruption in restructuring knowledge production factors. The convex adjustment cost reflects an increasing marginal cost of adjustment, and is quadratic in changes in R&D investment. To allow for zero R&D spending, the second term is quadratic in the midpoint percentage change (i.e., changes in investment relative to the average of the previous and current expenditures  $\frac{2(d_t - d_{t-1})}{d_t + d_{t-1}}$ ), rather than changes in rates (i.e.,  $\frac{d_t}{d_{t-1}} - 1$ ) as employed by Christiano et al. (2005) and Schmitt-Grohe and Uribe (2005). Otherwise, there would be an infinite cost to adjust from zero R&D and zero R&D would be an absorbing state.

The firm's future productivity level  $z_{t+1}$  is determined by its total knowledge input  $D_t$ , innovation uncertainty parameters  $\delta$  and  $\rho$  which will be defined below, and current status  $z_t$ , where the current productivity level  $z_t$  is a stochastic realization dependent on past productivity, R&D effort, and knowledge spillover. The transition in productivity from  $z_t$  to  $z_{t+1}$  is specified as follows:

$$\Gamma(z_{t+1}|z_t) = \begin{cases} \bar{z} \geq z_{t+1} > z_t & \text{with probability } \Pr(z_{t+1} > z_t) \times \Pr(z_{t+1}|z_{t+1} > z_t), \\ z_{t+1} = z_t & \text{with probability } \Pr(z_{t+1} = z_t), \\ \underline{z} \leq z_{t+1} < z_t & \text{with probability } \Pr(z_{t+1} < z_t) \times \Pr(z_{t+1}|z_{t+1} < z_t), \end{cases} \quad (5)$$

where

$$\Pr(z_{t+1}|z_{t+1} > z_t) = \frac{\frac{1}{(z_{t+1}-z_t)^\rho}}{\sum_{z_{t+1} > z_t} \frac{1}{(z_{t+1}-z_t)^\rho}}, \quad (6)$$

<sup>5</sup> The details of the derivation are in the appendix.

and

$$\Pr(z_{t+1}|z_t) = \frac{1}{\sum_{z_{t+1} < z_t} \frac{1}{(z_t - z_{t+1})^\rho}}. \quad (7)$$

Specifically, the probability of transitioning from current level  $z_t$  to future level  $z_{t+1}$  is equal to the product of the probability of directional movement and the probability of reaching  $z_{t+1}$  conditional on the directional changes. We follow Xu (2008) to specify the probability of directional movement. We assume that the firm improves its relative technological position and reaches a higher level with probability  $\Pr(z_{t+1} > z_t) = \frac{(1-\delta)D_t}{1+D_t}$ , maintains its past technological position with probability  $\Pr(z_{t+1} = z_t) = \frac{1-\delta+D_t}{1+D_t}$ , and loses its position and falls behind with probability  $\Pr(z_{t+1} < z_t) = \frac{\delta}{1+D_t}$ . The parameter  $\delta$  captures three knowledge-related risks: knowledge obsolescence or depreciation, firms' inability to keep up with competitors, and the uncertainty associated with innovation output. This specification also implies that higher investment in R&D improves firms' relative competitive positions by raising the likelihood of technological advancement.

Previous studies restrict firms from moving beyond the technology level that is closest to their current one, which is equivalent to the assumption that firms are only allowed to conduct incremental innovation and improve their technology gradually (Ericson and Pakes, 1995; Xu, 2008; Hashmi and Van Biesebroeck, 2016). Unlike those studies, we relax the assumption by incorporating radical innovation in the model. Incremental and radical innovations differ in several respects. In this paper, we focus on their differential effects on technology and revenue—that is, radical innovation improves technology more rapidly than incremental innovation and helps generate higher revenue. Specifically, we assume that firms' future technology  $z_{t+1}$  can move to any level, with the probability changing with its distance from current level  $z_t$ . The parameter  $\rho$  controls the likelihood of realizing extreme productivity. The lower the value of  $\rho$ , the more likely firms will realize radical innovation and high income. Moreover, the parameter  $\rho$  captures a wide range of possibilities. When  $\rho \rightarrow \infty$ , the firm will move to the value closest to  $z_t$  with near certainty, which is equivalent to the assumption made by previous studies. When  $\rho = 0$ , the firm will have an equal chance of reaching any level other than  $z_t$ , that is, an equal probability of realizing incremental innovation or radical innovation. When  $\rho \rightarrow -\infty$ , the firm will move to the farthest value of  $z_{t+1}$  with near certainty. A realistic value of  $\rho$  is between 0 and  $\infty$  where firms are more likely to move to closer productivity levels, and we will allow our estimation to pin down this parameter. Finally, we denote the conditional probability distribution for  $z_{t+1}$  by  $\Gamma(z_{t+1}|z_t)$ .

### 2.3. Financing

Firms can finance R&D investment through three different sources: current-period revenue, internal cash balance, and external financing.

Cash balance,  $c_t$ , earns a risk-free rate  $r$ . The amount of interest earned is taxed at the corporate tax rate  $\tau_c$  when taxable income is positive.

External financing takes the form of equity issuance. This assumption follows the observation by Brown et al. (2009) that equity finance is a more relevant substitute for internal cash flow because of the intangible nature and low collateral value of R&D.<sup>6</sup> Issuing equity incurs both fixed costs,  $\lambda_0$ , and linear costs,  $\lambda_1$  (Hennessy and Whited, 2007; Nikolov and Whited, 2014). Following previous studies, we describe negative equity flow  $e_t < 0$  as equity issuance and positive equity flow  $e_t \geq 0$  as dividend distribution.

### 2.4. The firm's problem

#### 2.4.1. The incumbent's problem

Each period, after productivity  $z_t$  is realized, the firm decides whether to leave the market by weighing the expected benefits of continuing to operate and of exiting.

If the firm chooses to stay, it hires labor  $l_t$  at the wage rate  $w$  to produce output and sets a price  $p_t$  on its product. Its taxable income is

$$\pi_t = p_t y_t - w l_t - c_F + r c_t - d_t, \quad (8)$$

in which  $c_F$  is the fixed operational cost. Upon receiving its profit, the firm makes decisions about R&D investment and cash savings. If available internal resources are insufficient to cover those expenses, the firm issues equity; otherwise, it distributes dividends. Net equity flow is

<sup>6</sup> Also, we find that for high-tech firms during the 2000s, the median short-term debt-to-assets ratio and long-term debt-to-assets ratio are 0.005 and 0.005, respectively. In addition, the average propensity to save out of equity issuance is 0.557, and most of the cash ratio increase originates from equity issuance when we consider the four primary sources of funds (equity issuance, debt issuance, earnings, and capital sales). These statistics confirm innovative firms' high relative reliance on equity financing.

$$g_I(e_t) = -\mathbf{1}_{e_t < 0} \lambda_0 + (1 - \mathbf{1}_{e_t \geq 0} \tau_d + \mathbf{1}_{e_t < 0} \lambda_1) e_t(\pi_t, c_t, c_{t+1}, d_{t-1}, d_t), \quad (9)$$

where gross equity flow before the dividend distribution tax and equity issuance cost is given by

$$e_t(\pi_t, c_t, c_{t+1}, d_{t-1}, d_t) = (1 - \tau_c \mathbf{1}_{\pi_t \geq 0}) \pi_t + (c_t - c_{t+1}) + \tau_{rd} d_t - A(d_{t-1}, d_t). \quad (10)$$

The first term of  $e_t$  is the firm's net income, and the other terms are the net change in cash, R&D subsidy, and adjustment costs, respectively. Parameters  $\tau_c$  and  $\tau_{rd}$  are the corporate income tax rate and R&D tax-credit rate. The indicator function  $\mathbf{1}_{e_t \geq 0}$  equals one if the firm distributes dividends and zero otherwise, while  $\mathbf{1}_{e_t < 0}$  equals one if the firm issues equity and zero otherwise. On the other hand, the indicator function  $\mathbf{1}_{\pi_t \geq 0}$  means that the firm is only taxed when taxable income is positive.

If the firm chooses to exit, the corresponding net equity flow is

$$g_X(e_t) = -\mathbf{1}_{e_t < 0} \lambda_0 + (1 - \mathbf{1}_{e_t \geq 0} \tau_d + \mathbf{1}_{e_t < 0} \lambda_1) e_t(\pi_t, c_t, c_{t+1} = 0, d_{t-1}, d_t = 0). \quad (11)$$

The risk-neutral incumbent maximizes its value. Moving on to the recursive formulation, let  $V_I(z, c, d_{-1}; \mu)$  and  $V_X(z, c, d_{-1}; \mu)$  denote the value functions of continuing to operate and exiting, respectively. Here, subscript  $-1$  indicates a variable in the preceding period. The variable  $\mu$  is a distribution which contains the idiosyncratic productivity levels, cash balances, and previous R&D spending of all firms, and this distribution will be specified in Subsection 2.5. Also, let  $x'$  denote the continue/exit decision where  $x' = 0$  indicates that the firm continues and  $x' = 1$  indicates that the firm exits. Then the incumbent's problem becomes

$$V(z, c, d_{-1}; \mu) = \max_{x' \in \{0,1\}} \{V_I(z, c, d_{-1}; \mu), V_X(z, c, d_{-1}; \mu)\}, \quad (12)$$

where

$$V_I(z, c, d_{-1}; \mu) = \max_{c', d} \{g_I(e) + \beta \mathbb{E} V(z', c', d; \mu')\}, \quad (13)$$

and

$$V_X(z, c, d_{-1}; \mu) = g_X(e). \quad (14)$$

The parameter  $\beta$  is the discount factor which equals  $\frac{1}{1+r}$ , and a prime indicates a variable in the subsequent period.

Note that frictions interact with each other. Equity issuance costs and the dividend tax ensure that the firm behaves like a risk-averse agent. Without financial frictions and the dividend tax, the firm's periodic utility function is  $g(e) = e$ , where  $e \geq 0$  is dividend payment and  $e < 0$  is equity issuance. This periodic utility function is linear. Therefore when  $\lambda_0 = \lambda_1 = \tau_d = 0$ , the firm is risk-neutral and demands zero precautionary cash regardless of uncertainty levels. With financial frictions and/or the dividend tax, the firm's periodic utility function becomes concave, and it acts as a risk-averse agent and has incentive to hold cash to hedge against uncertainty. For example, it is especially costly for a firm to distribute dividends but then find out that it actually needs to recall the funds as equity issuance in a future period. The other R&D frictions we consider in the paper—R&D adjustment costs, innovation uncertainty, knowledge spillover, and market competition—essentially influence the degree of uncertainty in a firm's spending (uses of funds) and/or income (sources of funds). These factors, together with financial frictions and the dividend tax, drive precautionary cash demand.

#### 2.4.2. The potential entrant's problem

A potential entrant pays a cost,  $c_E$ , at the beginning of a period to draw a  $z$  with probability

$$\Pr(z > \underline{z}) = \frac{\frac{1}{(z-\underline{z})^\rho}}{\sum_{z \geq \underline{z}} \frac{1}{(z-\underline{z})^\rho}}, \quad (15)$$

where  $\underline{z}$  is the lowest productivity level. This probability specification is the same as the one that incumbents face, and its corresponding probability distribution is denoted as  $\Gamma_E(z)$ .

Upon seeing  $z$ , the potential entrant can either choose to throw away the draw, or commit to enter the economy and produce  $y$ . If the potential entrant commits to enter, it carries  $c = 0$  and  $d_{-1} = 0$  from the previous period because it was not operational, and chooses  $c'$  and  $d$  to maximize the expected discounted value of the firm.

With this form of entry, the potential entrant throws away low- $z$  draws because potential profits are low and it does not want to pay the adjustment costs. On the other hand, high- $z$  draws induce entry in order to capture more substantial profits. If a firm enters, it also invests in cash and R&D to set itself up for the future.

The free-entry condition is

$$\int \max\{V_I(z, 0, 0; \mu), 0\} d\Gamma_E(z) \leq c_E, \quad (16)$$

and the condition holds with equality when there is a non-zero mass of entry.



### 2.5. Law of motion for distribution $\mu$

Conditional on the current-period joint distribution  $\mu(z, c, d_{-1})$  of idiosyncratic productivity levels, cash balances, and previous R&D spending, the next-period distribution is determined by

$$\begin{aligned} \mu'(z', c', d) = & \int I(z, c, d_{-1}; \mu) d\Gamma(z'|z) d\mu(z, c, d_{-1}) \\ & + M' \int I(z, 0, 0; \mu) d\Gamma(z'|z) d\Gamma_E(z), \end{aligned} \quad (17)$$

where  $I(z, c, d_{-1}; \mu) \equiv \mathbf{1}_{\mathcal{C}(z, c, d_{-1}; \mu)=c'} \mathbf{1}_{\mathcal{D}(z, c, d_{-1}; \mu)=d} \mathbf{1}_{\mathcal{X}(z, c, d_{-1}; \mu)=0}$  is a combined indicator function for incumbents,  $I(z, 0, 0; \mu) \equiv \mathbf{1}_{\mathcal{C}(z, 0, 0; \mu)=c'} \mathbf{1}_{\mathcal{D}(z, 0, 0; \mu)=d} \mathbf{1}_{\mathcal{X}(z, 0, 0; \mu)=0}$  is a combined indicator function for potential entrants, and  $M'$  is the mass of potential entrants. Note that  $\mathcal{C}(z, c, d_{-1}; \mu)$ ,  $\mathcal{D}(z, c, d_{-1}; \mu)$ , and  $\mathcal{X}(z, c, d_{-1}; \mu)$  are the cash, R&D, and exit decision rules, respectively, for incumbents, while  $\mathcal{C}(z, 0, 0; \mu)$ ,  $\mathcal{D}(z, 0, 0; \mu)$ , and  $\mathcal{X}(z, 0, 0; \mu)$  are the cash, R&D, and exit decision rules, respectively, for potential entrants. The “exit” decision  $\mathcal{X}(z, 0, 0; \mu) = 1$  for potential entrants means that they throw away the  $z$  draws, while the “continue” decision  $\mathcal{X}(z, 0, 0; \mu) = 0$  means that they enter and produce with their  $z$  draws.

The entry and exit dynamics in this type of model can be best described as “bottom churning.” That is, the worst firms exit every period and are replaced by entrants with higher  $z$  draws.

### 2.6. Industry equilibrium

In this paper, we focus on the stationary industry equilibrium.

**Definition 1.** A **stationary industry equilibrium** is a stationary distribution  $\mu$ , a price  $P$ , a quantity  $Y$ , and policy functions  $\mathcal{C}(z, c, d_{-1}; \mu)$ ,  $\mathcal{D}(z, c, d_{-1}; \mu)$ , and  $\mathcal{X}(z, c, d_{-1}; \mu)$  such that:

- (i) policy functions solve the firm’s problem, given industry price  $P$ , industry output  $Y$  and distribution  $\mu$ ;
- (ii) the distribution  $\mu$  is invariant over time;
- (iii) the free-entry condition is satisfied; and
- (iv) the product market clears.

## 3. Model estimation

In this section, we apply the model described in Section 2 to data. We first select and construct a set of data moments, using a sample of firms that (i) operate in 3-digit SIC high-tech industries which were determined by Brown and Petersen (2011)—SIC 283, SIC 357, SIC 366, SIC 367, SIC 382, SIC 384, and SIC 737—and (ii) have positive sales, positive assets, and non-missing R&D investment for the period 2000–2014 from Compustat. Our sample consists of 26,336 firm-year observations which implies an average of 1,756 firms per year. We then parameterize the model by matching selected data moments and show that our model is able to replicate high-tech firms’ R&D and cash-saving behavior observed in the data.

### 3.1. Parameterization

We set the length of a period in the model to be one year. The average real risk-free interest rate during the 2000s was 3%. We therefore set  $r$  to be 3%.

We discretize technology  $z$  to 10 levels and calibrate their values as follows. Using the methodology in İmrohoroglu and Tüzel (2014), firms’ value-added is calculated as the sum of their operating income before depreciation (item *oibdp*) and labor expenses and subsequently deflated by the production price index. Labor expenses are measured as the sum of employment payroll and stock-based compensation expenses (item *xintopt*), where employment payroll is the product of the number of employees (item *emp*) and the average wage in each 3-digit SIC industry calculated from the NBER-CES Manufacturing Industry Database. Given our production function specification ( $y_i = z_i l_i$ ), the technology level is the ratio of firms’ real value added to employment. We then normalize the mean technology level to be 1 and set the value of  $z_i$ , where  $i = \{1, 2, \dots, 10\}$ , as the median of each decile of the distribution.

We choose the elasticity of substitution between goods  $\sigma$  to match the markup charged by high-tech firms during the sample period. Markup is defined as the percentage of sales over cost of goods sold. We set  $\sigma$  to be 2 to match a 100% markup, which falls between the median and mean value suggested by our sample.

We calibrate dividend tax, corporate income tax, and R&D tax credits as follows. Dividend tax is chosen to be 15%, a rate commonly used in the literature (Hennessy and Whited, 2005). Corporate income tax  $\tau_c$  is set at 35%, which is the statutory corporate income tax in the 2000s. The Internal Revenue Code provides three methods to calculate R&D tax credits, the simplest of which is called the Alternative Simplified Credit method. Using this approach, firms can claim a tax credit equal to 14% of qualified R&D expenditures that exceed a calculated base amount, while the base amount cannot be less than 50% of their current-year qualified expenditures. Therefore, we set the R&D tax credits to be 6%, slightly lower than half the 14% rate.

**Table 1**

**Model Parameterizations.** This table summarizes the parameters used to solve the model at annual frequency. Panel A reports the parameters calibrated separately from data or borrowed from other studies. Panel B presents estimation results by taking parameters in Panel A as given and jointly matching twelve selected data moments. Standard errors are presented in parentheses.

| Panel A: Parameters Calibrated Separately        |        |          |
|--------------------------------------------------|--------|----------|
| Real risk-free rate ( $r$ )                      | 0.03   |          |
| Elasticity of substitution ( $\sigma$ )          | 2.00   |          |
| Dividend tax ( $\tau_d$ )                        | 0.15   |          |
| Statutory corporate income tax ( $\tau_c$ )      | 0.35   |          |
| R&D tax credits ( $\tau_{rd}$ )                  | 0.06   |          |
| Panel B: Parameters Estimated by SMM             |        |          |
| Fixed costs of external finance ( $\lambda_0$ )  | 0.0453 | (0.0055) |
| Linear costs of external finance ( $\lambda_1$ ) | 0.1251 | (0.0256) |
| Fixed adjustment cost ( $\gamma_0$ )             | 0.0643 | (0.0153) |
| Convex adjustment cost ( $\gamma_2$ )            | 2.3799 | (0.0479) |
| Innovation efficiency ( $\delta$ )               | 0.1948 | (0.0001) |
| Knowledge spillover ( $\theta$ )                 | 0.5930 | (0.0001) |
| Innovation volatility ( $\rho$ )                 | 0.7573 | (0.0014) |
| Fixed operating cost ( $c_F$ )                   | 0.6596 | (0.0001) |
| Wage cost ( $w$ )                                | 0.2248 | (0.0035) |

Parameters are shown in Table 1. Panel A summarizes the parameters that are either directly calibrated from data or borrowed from other relevant studies. Panel B reports the remaining parameters, which are estimated using a Simulated Method of Moments (SMM) approach. That is, we recover underlying parameters from a list of selected moments by minimizing the distance between the moments constructed from model-simulated data and the moments computed with actual data. The match is quite close and the details are discussed in Subsection 3.2.2.

### 3.2. Comparative statics and parameter identification

To identify the parameters that are not predetermined, we need moments that are informative—that is, sensitive to parameter changes. To this end and also to provide intuition about how each R&D feature influences high-tech firms' cash and R&D decisions, we report comparative statics of the following parameters: R&D adjustment costs ( $\gamma_0$  and  $\gamma_2$ ), innovation uncertainty ( $\delta$  and  $\rho$ ), knowledge spillover ( $\theta$ ), external financing costs ( $\lambda_0$  and  $\lambda_1$ ), and market competition ( $\sigma$  and  $c_F$ ). These parameters take the values of equally spaced points in the following intervals, respectively:  $\gamma_0 \in [0, 0.1]$ ,  $\gamma_2 \in [1, 3]$ ,  $\delta \in [0.1, 0.3]$ ,  $\rho \in [0, 2]$ ,  $\theta \in [0, 1.5]$ ,  $\lambda_0 \in [0, 0.1]$ ,  $\lambda_1 \in [0, 0.3]$ ,  $\sigma \in [1.9, 2.1]$ , and  $c_F \in [0, 1]$ . These intervals represent reasonable bounds on each parameter since the estimated parameter values are close to the center point of the corresponding intervals. We change one parameter at a time, holding all other parameters at the calibrated or estimated values reported in Table 1.

#### 3.2.1. Comparative statics

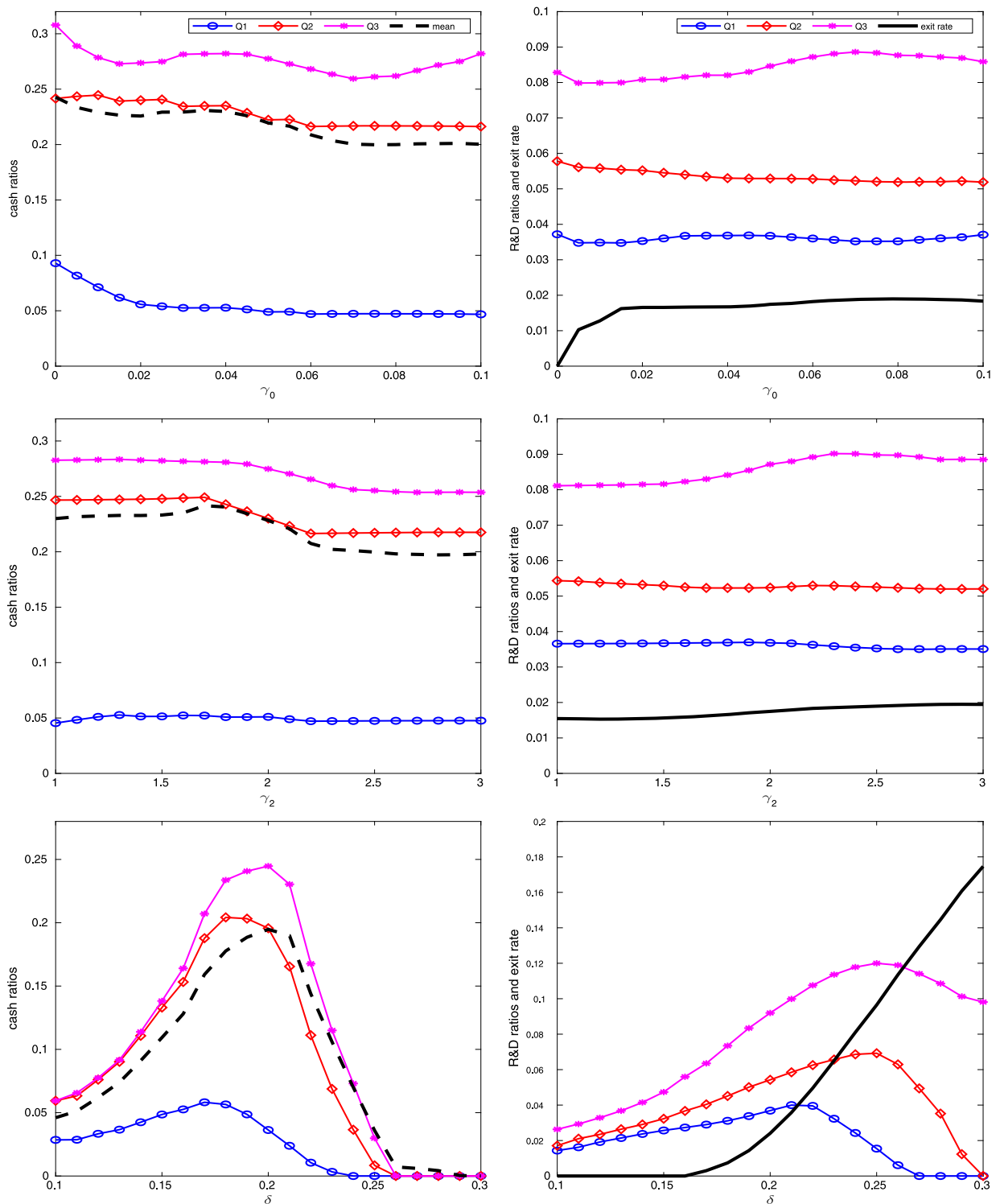
Figs. 1–3 plot the comparative statics results. Panels in the left column show the sensitivity of the first quartile, second quartile, third quartile, and industry mean of cash-to-firm value ratios with respect to each parameter, and panels on the right present the sensitivity of the first quartile, second quartile, third quartile of R&D-to-firm value ratios and exit rates.<sup>7</sup> We report exit rates to give information about the changes in the stationary distribution of firms.

**3.2.1.1. R&D adjustment costs  $\gamma_0$  and  $\gamma_2$**  Brown and Petersen (2011) argue that adjusting R&D spending is costly, and therefore firms facing financial frictions tend to accumulate valuable internal funds to smooth their R&D investment and avoid potentially large adjustment costs. More specifically, a large share of R&D expenses goes to pay the salaries of highly educated scientists, engineers, and other specialists. Their effort creates a firm's knowledge base, which is embedded in its human capital. Discovering and integrating these talented workers can be costly. On the other hand, turnover among these skilled workers can erode the firm's knowledge base and lead to dissemination of proprietary information. Firms, therefore, have strong incentives to retain skilled employees, as reflected by the highly persistent R&D spending—which, in turn, implies substantial adjustment costs.

The effects of R&D adjustment costs ( $\gamma_0$  and  $\gamma_2$ ) are shown in the top four panels of Fig. 1. An increase in  $\gamma_0$  raises investment costs uniformly for firms, regardless of the magnitude of their investment adjustment, while a rise in  $\gamma_2$  increases

<sup>7</sup> To limit the effects of extreme outliers generated by sales data, we examine R&D-to-firm value and cash-to-firm value ratios, instead of R&D-to-sales and cash-to-sales ratios. Note that we do not report mean R&D-to-firm value ratios because the normalization still produces many outliers in the data and in the model. There are some firms with close to zero firm value that perform significant amounts of R&D. Accordingly, we do not target this moment in the estimation. Cash does not suffer from the same issue since it adds directly to firm value.





**Fig. 1. Comparative Statics I.** This figure plots the effects of fixed R&D adjustment costs  $\gamma_0$ , convex R&D adjustment costs  $\gamma_2$ , and innovation efficiency  $\delta$  on (i) cash-to-firm value ratios, (ii) R&D-to-firm value ratios, and (iii) exit rates.

the investment costs proportional to the square of adjustment. Both costs place a heavier financial burden on firms and absorb cash flow which otherwise would be directed to cash holdings.

However, quantitatively, these effects are rather minor. R&D investment tends to be naturally quite persistent, and R&D adjustment costs therefore do not have a large impact on cash policy. In particular, raising or lowering R&D investment does not have a direct effect on productivity level, and accordingly, firms change R&D levels only when necessary. This implies that higher adjustment costs raise the cost of investment moderately and predictably without strengthening the precautionary motive much. Instead, firms tend to draw down cash holdings to meet higher adjustment related expenditures.

**3.2.1.2. Innovation efficiency  $\delta$**  The parameter  $\delta$  captures innovation efficiency, which affects the marginal product of R&D investment (Lyandres and Palazzo, 2014; Hsu et al., 2016). Hsu et al. (2016) suggest a positive role of innovation efficiency in shaping high-tech firms' cash-saving policies. They argue that an increase in innovative efficiency (i.e., a drop in  $\delta$ ) generates stronger precautionary considerations by raising the likelihood of innovation success as well as the opportunity costs of being financially distressed.

The responses of cash and R&D ratios to changes in  $\delta$  are reported in the bottom two panels of Fig. 1. Overall, firms at different levels of R&D intensity react to changes in  $\delta$  in a similar fashion. An increase in  $\delta$  makes innovation less efficient. Facing a higher probability of innovation failure, firms decide to invest more in R&D so as to improve innovation success rates and compete in the market. This explains the upward trend of the R&D ratio. However, as  $\delta$  further increases, fewer firms can survive in such uncertain environments. Followers that decide to stay choose to take advantage of knowledge spillover and reduce their R&D investment to zero, while leaders struggle to maintain their R&D spending. This explains the decline in the R&D ratio when  $\delta$  is high.

An increase in  $\delta$  affects firms' saving decisions through three interesting channels. First is the precautionary consideration. Higher R&D spending generates stronger demand for precautionary savings. Second is the substitution effect. Shifting more resources to R&D implies that less is left for cash savings. Third is the income effect. A lower chance of innovation success reduces cash flow that can be allocated to cash stocks. When  $\delta$  is small, the first channel dominates: a higher  $\delta$  leads to an increase in cash holdings. As  $\delta$  continues to increase, the latter two channels become important, and cash ratios start to decrease with  $\delta$ .

**3.2.1.3. Innovation volatility  $\rho$**  Incremental and radical innovations have different impacts on firms' competitive positions and revenue. Little attention so far has been paid to the effect of this feature on cash policies. As shown in the top two panels of Fig. 2, a rise in  $\rho$ —that is, a decline in the probability of realizing extreme outcomes—plays a role similar to an increase in  $\delta$ , except for its effects on R&D spending.

In response to an increase in  $\rho$  which implies a lower chance of drawing extremely negative and positive productivity, all firms become less concerned about radical failure and choose to increase their R&D expenditures to improve the likelihood of realizing radical innovation, yet in different magnitudes. High R&D intensive firms are the ones that respond more aggressively.

An increase in  $\rho$  impacts firms' cash decisions in a way similar to  $\delta$ . First, it strengthens firms' incentives to invest in R&D, which increases firms' needs for precautionary cash. Second, it shifts more resources to R&D, which reduces the resources available for cash reserves. These two channels compete with each other. When  $\rho$  is high, the second channel dominates.

**3.2.1.4. Knowledge spillover  $\theta$**  Qiu and Wan (2015) emphasize a positive effect of technology spillover on cash policies. They argue that the enhancement of the marginal product of R&D via spillover provides firms with stronger incentives to apply external knowledge and perform R&D, which in turn leads to a higher demand for cash.

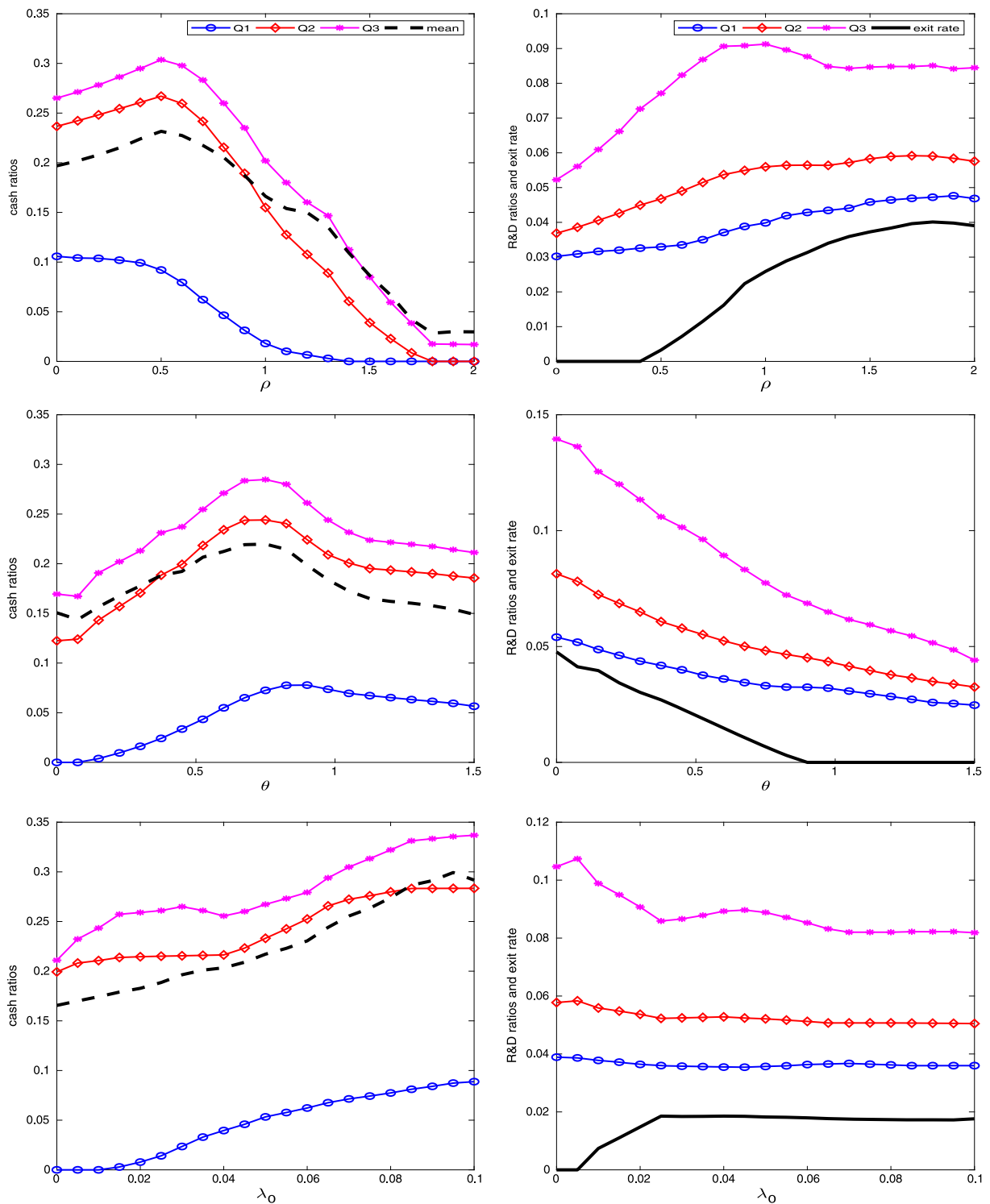
Results of the effect of spillover on cash holdings are reported in the middle two panels of Fig. 2. They are in line with the empirical findings by Qiu and Wan (2015), yet suggest a different mechanism. As spillover becomes greater, R&D intensive firms effectively have lower returns to knowledge investment. They are discouraged from innovating and choose to cut their R&D spending. Meanwhile, followers free ride leaders and reduce their own R&D expenditures.

Lower R&D investment weakens the precautionary motive for cash. However, the decrease in precautionary cash savings is overpowered by the increase in the substitution away from R&D to cash (substitution effect) and the increase in income that knowledge spillover brings (income effect). Therefore, an increase in knowledge spillover drives up cash ratios until the parameter is larger than 0.8.

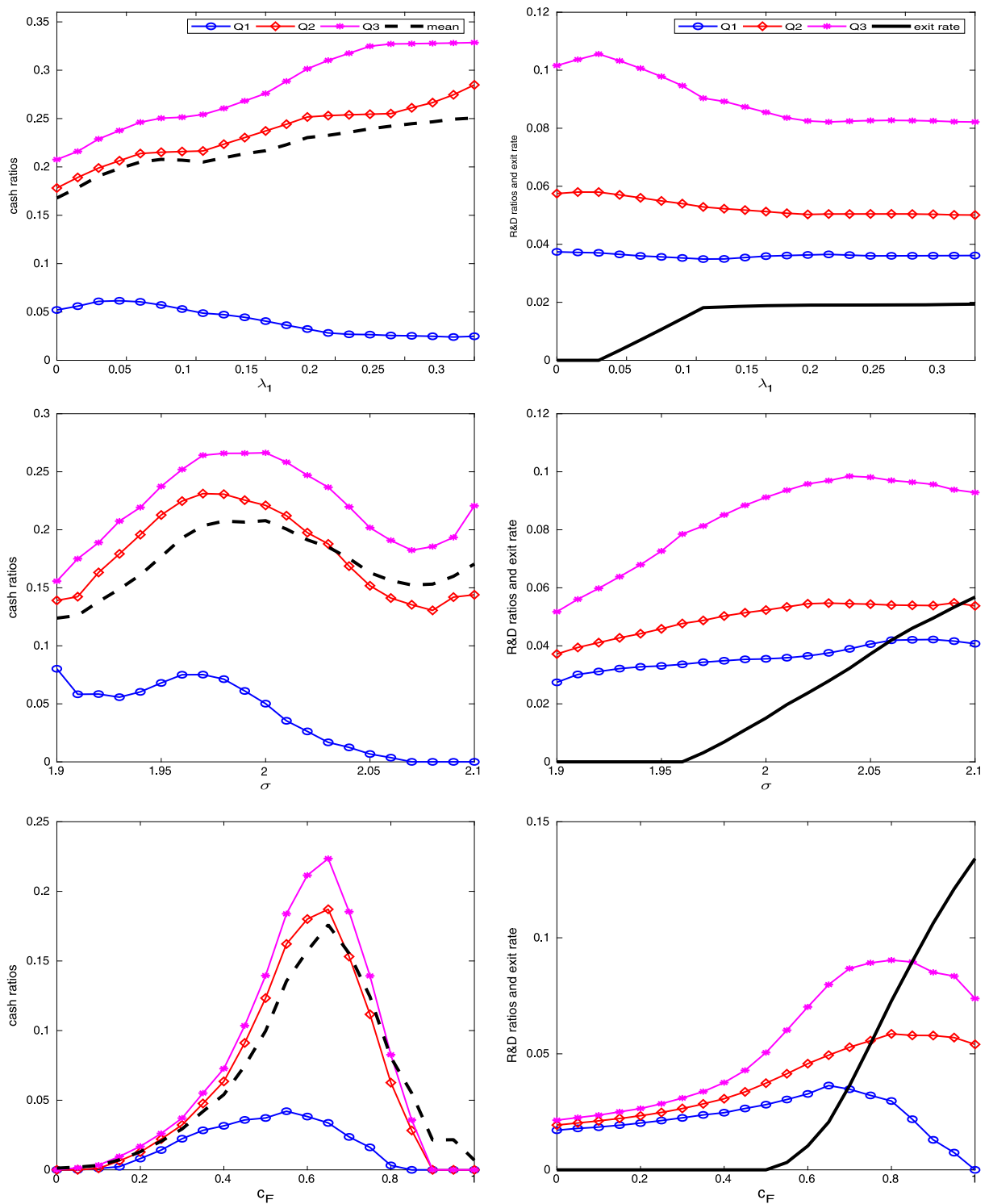
A common theme so far is that the substitution and income effects are often at odds with the precautionary motive to hold cash. These conflicting forces then shape the cash holding behavior of firms. In this case, the non-monotonicity is not large given the wide range of  $\theta$  in the comparative statics exercise.

**3.2.1.5. Financial frictions  $\lambda_0$  and  $\lambda_1$**  One of the features frequently used to explain high-tech firms' large cash positions is financial frictions (Falato et al., 2013; Falato and Sim, 2014; Hall and Lerner, 2010). Due to information asymmetries and low collateral value of R&D, high-tech firms face expensive equity-issuance costs. To avoid underinvestment in R&D caused by financial constraints, firms accumulate internal funds.

Our results, reported in the bottom two panels of Fig. 2 and the top two panels of Fig. 3, suggest the following: An increase in financing costs,  $\lambda_0$  and  $\lambda_1$ , has no significant adverse effects on R&D investment, except for firms with high R&D



**Fig. 2. Comparative Statics II.** This figure plots the effects of innovation volatility  $\rho$ , knowledge spillover  $\theta$ , and fixed equity issuance costs  $\lambda_0$  on (i) cash-to-firm value ratios, (ii) R&D-to-firm value ratios, and (iii) exit rates.



**Fig. 3. Comparative Statics III.** This figure plots the effects of linear equity issuance costs  $\lambda_1$ , elasticity of substitution  $\sigma$ , and fixed operating costs  $c_F$  on (i) cash-to-firm value ratios, (ii) R&D-to-firm value ratios, and (iii) exit rates.

that choose to cut their R&D ratio by about 0.02. This finding implies that firms' internal cash reserves largely eliminate possible underinvestment in R&D caused by financial frictions. At a certain point, further increases in financing costs do not have an impact on firms since they choose to self finance.

In addition, fixed and linear equity issuance costs have different impacts on the cash policies of cash-poor firms. These firms choose to increase their cash ratios in response to an increase in  $\lambda_0$  to cut down on fixed issuance costs. On the other hand, they reduce their cash holdings if  $\lambda_1$  increases. When they face worse credit market conditions due to a proportional increase in external-financing costs, cash-poor firms choose to issue equity in smaller amounts and save less.

**3.2.1.6. Market competition  $\sigma$  and  $c_F$**  Market competition is another often-discussed determinant of high-tech firms' cash policies (He and Wintoki, 2016; Lyandres and Palazzo, 2016; Ma et al., 2020). It can be proxied by two parameters in our model: elasticity of substitution between differentiated goods ( $\sigma$ ) and fixed operating costs ( $c_F$ ), with comparative statics shown in the last four panels of Fig. 3. Admittedly, our measure of market competition is rather reduced form. However, we can still capture the fact that increased competition reduces profits without severely complicating the model.

Elasticity of substitution  $\sigma$  reflects the degree of responsiveness in demand with respect to price changes and determines the markup  $\left(\frac{\sigma}{\sigma-1}\right)$  that firms can charge above their marginal costs ( $w/z$ ). An increase in the elasticity  $\sigma$  intensifies product market competition, and reduces markup and the expected benefit of entry. It drives up exit rates and induces higher-productivity firms to enter, which further increases competitive pressures in the industry. Firms therefore raise their R&D spending to compete and demand more precautionary cash savings. However, higher R&D investment implies higher expenditures, which compete for resources with cash savings. The second channel dominates the first at high values of  $\sigma$  and generates a hump-shaped response in cash ratios.

Compared to an increase in  $\sigma$ , an increase in operating costs  $c_F$ —another measure of intensified market competition—has qualitatively similar but quantitatively different impacts on high-tech firms' cash and R&D policies. The graphs pertaining to  $c_F$  illustrate that cash and R&D drop significantly when market exit rates rise up to 10% and higher.

Also, when  $\sigma$  approaches 2.1, the average cash ratio, primarily driven by firms with high cash holdings, starts to rise again. We conjecture that this increase arises from strong income effects. That is, an increase in  $\sigma$  raises firm exit rates. Incumbents' market share and revenue thus increase without a substantial change in their cost structure. Incumbents then allocate the additional revenue to cash holdings. This same mechanism does not hold for an increase in  $c_F$ , because operating costs outpace the revenue generated from higher market shares. Therefore, disentangling different sources of market competition is necessary and can have important implications for the choice of pro-competition policies.

### 3.2.2. Parameter identification

Given the comparative statics in the previous subsection as well as from additional analyses, we choose the following moments to jointly estimate our undetermined model parameters: the 25<sup>th</sup> and 75<sup>th</sup> percentiles of R&D-to-firm value and cash-to-firm value ratios, serial correlation of within-firm R&D and cash ratios, their correlation, the fraction of firms with zero R&D investment, the mean, standard deviation and serial correlation of productivity, and firm exit rate.

In particular, in the face of high fixed costs of adjusting R&D expenses  $\gamma_0$ , firms may choose to be non-R&D performers. The fraction of firms with zero R&D investment therefore can be used to infer  $\gamma_0$ . Given the degree of innovation uncertainty, the persistence of within-firm R&D ratio provides information on the magnitude of convex R&D adjustment costs  $\gamma_2$ .

The persistence of within-firm cash ratios can be used to infer the magnitude of fixed equity-issuance costs  $\lambda_0$ . A smaller  $\lambda_0$  leads to less lumpy equity issuance and a more smooth cash ratio. Cash-poor firms tend to be more sensitive to changes in credit market conditions. The 25<sup>th</sup> percentile of the cash ratio thus contains information about linear equity-issuance costs  $\lambda_1$ .

An increase in  $\delta$  has significant impacts on firms' R&D effort. It implies a higher chance of innovation failure and induces firms to invest more in R&D in order to promote success rates. We therefore use the 25<sup>th</sup> percentile of the R&D ratio to identify  $\delta$ . Changes in  $\rho$  affect the prospects for realizing breakthroughs, which has impacts on leaders' income and R&D effort and in turn affects their cash holdings through both substitution and income effects. We thus use the 75<sup>th</sup> percentile of the cash ratio to estimate  $\rho$ . An increase in  $\theta$  tends to have a discouraging effect on R&D investments. The 75<sup>th</sup> percentile of the R&D ratio, therefore, is informative about the size of knowledge spillover  $\theta$ .

We infer fixed operating costs  $c_F$  from firm exit rates. As  $c_F$  increases, fewer firms can cover the costs and are more likely to choose to exit the market. We estimate wage costs  $w$  from the correlation between R&D investment and cash holdings. In response to a higher wage rate, firms pass the increased production costs to customers by setting higher product prices. The corresponding changes in income move cash and R&D investment in the same direction through the income effect, which generates a higher correlation between them.

We also include productivity-related moments as additional information to further discipline our estimation.

### 3.3. Estimation results

In this subsection, we report estimation results of the model described in Section 2.

### 3.3.1. Parameter estimates

Parameter estimates are reported in Panel B of Table 1. The estimated  $\gamma_0$  is 0.064, which implies that the fixed R&D adjustment cost is roughly equal to 2.3% of an average firm's sales. The estimated  $\gamma_2$  is 2.38. This value suggests that if current R&D investment deviates from the past value by 10%, the convex adjustment costs would amount to approximately 1.1% of R&D investment. Very few structural estimates of R&D adjustment costs are available in the literature. Li and Liu (2012) find the convex adjustment cost is larger than that of physical capital; their estimates range from 3.26 to 67.47, depending on model specifications. Kung (2015) sets the convex R&D adjustment cost parameter to be 3.3 by matching the relative volatility of R&D investment growth to consumption growth in the data. Our estimate of the convex adjustment costs  $\gamma_2$  appears to be close to the lower bound of their estimates. Given the different adjustment-cost specifications, we should be cautious about making direct comparisons. However, our estimate delivers a message consistent with that of Li and Liu (2012) and other R&D studies—that is, R&D adjustment costs are high.

Two other parameters inferred from firms' R&D and cash behavior are innovation efficiency  $\delta$  and knowledge spillover  $\theta$ . The estimate of  $\delta$  is 0.195, which implies moderate risks associated with innovation in general. The estimate of  $\theta$  is critical for measuring returns to R&D for policy considerations. Our estimate of  $\theta$  is 0.593, which is lower than the estimate provided by Xu (2008) and indicates relatively small intra-industry externalities among publicly traded technology companies.

The estimated fixed and linear equity-issuance costs,  $\lambda_0$  and  $\lambda_1$ , are 0.045 and 0.125, respectively. The linear cost is higher than the value estimated by Hennessy and Whited (2007) who find it to be approximately 0.091. We conjecture that our higher estimate arises from a different sample of firms. We examine high-tech companies that tend to be collateral poor due to information asymmetries and thus face higher external financing costs.

The parameter  $\rho$  is newly introduced to the R&D literature. It relaxes the restriction imposed by previous studies in which firms can only conduct incremental innovation and move to a productivity level that is equal or close to their current positions (Xu, 2008; Hashmi and Van Biesebroeck, 2016). In the classic productivity ladder without  $\rho$ , the autocorrelation of productivity would not be strongly affected by changes in jump probabilities, and there is no direct way to push a portion of the distribution to both tails. Therefore, it is difficult to have much of an effect on and disentangle the standard deviation and autocorrelation of productivity. Alternatively, in our model, R&D expenditures, the innovation efficiency parameter  $\delta$ , and the innovation volatility parameter  $\rho$  provide the delicate balance to match the productivity moments well. When  $\rho$  equals zero, there is an equal chance of realizing incremental and radical innovation and reaching any level other than the current one. The larger the parameter, the greater the likelihood that firms will move to the closest value. The estimated  $\rho$  is 0.757, implying a 4.9% chance of moving from median productivity to the technological frontier.

Lastly, the fixed operating cost  $c_F$  is 0.660. This value amounts to over 23% of an average firm's sales.

### 3.3.2. Simulated model moments

Tables 2 and 3 report the data moments and their corresponding simulated model moments, including R&D-to-firm value and cash-to-firm value ratios, share of non-R&D performers, productivity, and exit rates. To be consistent with our model, we treat it as an exit if firms are deleted from Compustat due to bankruptcy (item *dlrsn* is 2), liquidation (item *dlrsn* is 3), or acquisition and merger (item *dlrsn* is 1, 4, or 6) when previous sales growth is negative.

Table 2 presents the targeted model moments at the point estimates reported in Table 1. As shown, nearly all targets are well-matched. The model-implied first and third quartiles of R&D and cash distributions, within-firm serial correlation of R&D investment, correlation between R&D and cash choices, fraction of non-R&D performers, and exit rates are all close to data counterparts.

Although the first quartile of R&D is close in absolute terms, it has the largest deviation in percentage terms. In general, it is somewhat difficult for the model to match the extremely wide cross-sectional dispersion in R&D expenditures. The productivity moments are also a bit off but still quite good considering that we have an overidentified model.

We further evaluate model performance by comparing the model-implied distributions of R&D and cash ratios with the empirical distributions derived from data. We consider the following nontargeted moments: industry average, cross-sectional dispersion, industry median, and the  $k^{\text{th}}$  percentiles of the cash and R&D distributions with  $k = \{1, 5, 10, 90, 95, 99\}$ . Results are reported in Table 3. As evidenced, model-implied distributions closely resemble the data. The industry average, cross-sectional dispersion, and the right tails of R&D and cash distributions are all close to their data counterparts. An exception is the left tail of the cash-to-firm value ratio. Compared to the data, the model produces a slightly thinner left tail for the cash distribution—5<sup>th</sup> and 10<sup>th</sup> percentiles—and overshoots the industry median cash ratio, 0.135 vs. 0.217.

We also computed a few more statistics in order to understand the relationship between innovation, productivity, and firm size. Specifically, we computed the correlation between R&D and productivity and find a value of 0.281 for data firms and 0.271 for model firms. This match is extremely good and helps provide confidence that we link R&D and innovation in approximately the correct way.

However, firm size statistics do not do as well. We measure firm size using firm market value. Other measures of firm size are not ideal in our setting since sales are perfectly correlated with productivity as well as with labor because labor is statically optimized. We find that the correlation between R&D and firm size is 0.396 for model firms and 0.875 for data firms. This correlation is so high for Compustat firms because they span a vast size range. Finally, the correlation between productivity and firm size is 0.316 in the data and 0.938 in the model. Real world firms are of course very complex and our model cannot capture many other factors that affect firm size. Overall, the signs of the three correlations mentioned above are all consistent in the model and data.



**Table 2**

**Simulated Model Moments: Targeted Moments.** This table reports both data and corresponding model moments. Data moments are calculated based on a sample of Compustat high-tech firms over the period 2000–2014. The constructed weighting matrix minimizes mean squared percent deviations between data and model moments.

| Moments                                                    | Data  | Model |
|------------------------------------------------------------|-------|-------|
| (i) R&D:                                                   |       |       |
| 25 <sup>th</sup> percentile of R&D-to-firm value ratio     | 0.021 | 0.035 |
| 75 <sup>th</sup> percentile of R&D-to-firm value ratio     | 0.100 | 0.091 |
| Serial correlation of within-firm R&D-to-firm value ratio  | 0.518 | 0.516 |
| Fraction of firms with zero R&D investment                 | 0.021 | 0.017 |
| (ii) Cash:                                                 |       |       |
| 25 <sup>th</sup> percentile of cash-to-firm value ratio    | 0.061 | 0.047 |
| 75 <sup>th</sup> percentile of cash-to-firm value ratio    | 0.249 | 0.257 |
| Serial correlation of within-firm cash-to-firm value ratio | 0.693 | 0.588 |
| Correlation with R&D-to-firm value ratio                   | 0.369 | 0.326 |
| (iii) Productivity:                                        |       |       |
| Industry average productivity                              | 1.000 | 1.274 |
| Cross-sectional dispersion of productivity                 | 0.561 | 0.697 |
| Serial correlation of within-firm productivity             | 0.797 | 0.824 |
| (iv) Market competition:                                   |       |       |
| Exit rate                                                  | 0.019 | 0.019 |

**Table 3**

**Simulated Model Moments: Nontargeted Moments.** This table presents nontargeted moments for the R&D-to-firm value ratio and cash-to-firm value ratio. Data moments are calculated based on a sample of Compustat high-tech firms over the period 2000–2014.

| Moments                                                | Data  | Model |
|--------------------------------------------------------|-------|-------|
| (i) R&D:                                               |       |       |
| Industry average R&D-to-firm value ratio               | 0.081 | 0.088 |
| Cross-sectional dispersion of R&D-to-firm value ratio  | 0.109 | 0.122 |
| Industry median R&D-to-firm value ratio                | 0.048 | 0.053 |
| 1-percentile of R&D-to-firm value ratio                | 0.000 | 0.000 |
| 5-percentile of R&D-to-firm value ratio                | 0.003 | 0.032 |
| 10-percentile of R&D-to-firm value ratio               | 0.008 | 0.035 |
| 90-percentile of R&D-to-firm value ratio               | 0.181 | 0.168 |
| 95-percentile of R&D-to-firm value ratio               | 0.259 | 0.245 |
| 99-percentile of R&D-to-firm value ratio               | 0.517 | 0.624 |
| (ii) Cash:                                             |       |       |
| Industry average cash-to-firm value ratio              | 0.193 | 0.200 |
| Cross-sectional dispersion of cash-to-firm value ratio | 0.211 | 0.197 |
| Industry median cash-to-firm value ratio               | 0.135 | 0.217 |
| 1-percentile of cash-to-firm value ratio               | 0.000 | 0.000 |
| 5-percentile of cash-to-firm value ratio               | 0.008 | 0.000 |
| 10-percentile of cash-to-firm value ratio              | 0.021 | 0.000 |
| 90-percentile of cash-to-firm value ratio              | 0.424 | 0.374 |
| 95-percentile of cash-to-firm value ratio              | 0.580 | 0.514 |
| 99-percentile of cash-to-firm value ratio              | 1.006 | 0.976 |

We then investigate how firm age affects the negative equity flow (equity issuance) or positive equity flow (dividend distribution) level for both the model and data. We find that younger firms issue much more equity on average. Using Compustat, we determine that the average equity flow to firm value ratio is -0.091 for firms 5 years and younger, and -0.016 for firms older than 5 years. Equity flow is defined as dividend payment (item *dvc* plus *dvp*) plus equity repurchase (item *prstk*) minus equity issuance (item *sstk*). The average equity flow for model firms is -0.189 for young firms, and -0.028 for old firms where age is defined in the same way. The equity flow to firm value ratio is even negative for old firms because low value firms tend to issue equity regardless of age. If we look at unnormalized or raw equity flow, young firms issue equity on average while old firms distribute dividends—this is true for both data and model firms. Unnormalized values cannot be compared numerically but we find that the ratio of old firm average dividend distribution over young firm average equity issuance is 6.3 and 8.9 for data and model firms, respectively.

In addition, we looked at R&D investment differences for young and old firms. Young firms tend to invest less in R&D as a fraction of firm value. The R&D ratio is 0.078 and 0.082 for young and old Compustat firms, respectively. This difference in R&D investment is more pronounced for young and old model firms at 0.049 and 0.091, respectively. If we turn to raw

**Table 4**

**Local Role of R&D Features.** This table summarizes the effects of each key R&D feature on firms' R&D and cash choices locally, by changing the value of each parameter by 5% in the direction which enhances or strengthens the corresponding mechanism. The arrows beside each parameter indicate whether the parameter increases or decreases to strengthen the mechanism. The percentage change in average firm value compared to the benchmark model is reported in the last column.

|                                                             | Cash  |                          |                          | R&D    |                          |                          | Firm Value |
|-------------------------------------------------------------|-------|--------------------------|--------------------------|--------|--------------------------|--------------------------|------------|
|                                                             | Mean  | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | Median | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | % change   |
| <b>Benchmark Model</b>                                      | 0.200 | 0.047                    | 0.257                    | 0.053  | 0.035                    | 0.091                    |            |
| <b>R&amp;D Adjustment Costs</b>                             |       |                          |                          |        |                          |                          |            |
| Fixed adjustment cost ( $\gamma_0 \uparrow 0.0675$ )        | 0.200 | 0.047                    | 0.253                    | 0.052  | 0.035                    | 0.091                    | −0.4%      |
| Convex adjustment cost ( $\gamma_2 \uparrow 2.499$ )        | 0.200 | 0.047                    | 0.253                    | 0.053  | 0.035                    | 0.092                    | −0.9%      |
| <b>Innovation Uncertainty</b>                               |       |                          |                          |        |                          |                          |            |
| Innovation efficiency ( $\delta \downarrow 0.185$ )         | 0.229 | 0.086                    | 0.299                    | 0.047  | 0.032                    | 0.079                    | 20.0%      |
| Innovation volatility ( $\rho \downarrow 0.719$ )           | 0.248 | 0.064                    | 0.318                    | 0.050  | 0.035                    | 0.084                    | 7.0%       |
| <b>Externality</b>                                          |       |                          |                          |        |                          |                          |            |
| Knowledge spillover ( $\theta \uparrow 0.623$ )             | 0.240 | 0.064                    | 0.281                    | 0.050  | 0.035                    | 0.085                    | 5.92%      |
| <b>Financial Frictions</b>                                  |       |                          |                          |        |                          |                          |            |
| Fixed equity-issuance costs ( $\lambda_0 \uparrow 0.0476$ ) | 0.200 | 0.047                    | 0.253                    | 0.053  | 0.035                    | 0.092                    | −0.6%      |
| Linear equity-issuance cost ( $\lambda_1 \uparrow 0.131$ )  | 0.202 | 0.047                    | 0.261                    | 0.053  | 0.036                    | 0.091                    | 0.36%      |
| <b>Market Competition</b>                                   |       |                          |                          |        |                          |                          |            |
| Elasticity of substitution ( $\sigma \uparrow 2.100$ )      | 0.171 | 0.000                    | 0.221                    | 0.054  | 0.041                    | 0.093                    | −31.1%     |
| Fixed operating costs ( $c_F \uparrow 0.693$ )              | 0.175 | 0.016                    | 0.228                    | 0.057  | 0.039                    | 0.104                    | −14.8%     |

R&D, the ratio of mean young firm to mean old firm R&D investment is approximately 1:4 in the data, the corresponding model ratio, on the other hand, is approximately 1:3.

Overall, the model is able to achieve a good match for both cross-sectional heterogeneity and within-firm dynamics and explain the key patterns of firms' R&D and cash-holding behavior observed in the data. This validates and serves as a test of our model to strengthen the reliability of parameter estimates reported in Table 1.

#### 4. Model implications

In the following subsections, we use the parameter estimates obtained in Section 3 to quantitatively evaluate the role of each R&D feature in explaining the observed large cash stocks maintained by high-tech firms. We further use the model to shed light on the effectiveness of cash savings in mitigating R&D distortions from financial frictions. A robustness test of our model parameters and main model implications is also performed.

##### 4.1. Why do high-tech firms hold so much cash?

In this subsection, we rely on the structural estimates obtained above to quantify the importance of key R&D features in explaining innovative firms' cash and R&D choices. We consider the following factors: R&D adjustment costs ( $\gamma_0$  and  $\gamma_2$ ), innovation uncertainty ( $\delta$  and  $\rho$ ), knowledge spillover ( $\theta$ ), financial frictions caused by the non-collateralizability of R&D and information asymmetry between firms and investors ( $\lambda_0$  and  $\lambda_1$ ), and market competition ( $\sigma$  and  $c_F$ ). We change one parameter at a time, holding all other parameters constant. Three sets of perturbations are examined: (i) local effects by increasing the intensity of each mechanism by 5%, (ii) a robustness check by increasing the intensity of each mechanism such that mean firm value changes by 1%, and (iii) global effects by shutting down each channel.

##### 4.1.1. Local effects

First, we explore the effects of each feature locally by changing their parameter values by 5%. We change the parameters in the direction which enhances or strengthens each mechanism. For instance, lowering  $\rho$  increases innovation volatility while raising  $\theta$  increases knowledge spillover. Results are reported in Table 4.

Our estimation results suggest that R&D adjustment costs may not be a first-order factor in explaining innovative firms' high cash balances, which echoes the comparative statics analysis in Subsection 3.2.1.1. A 5% increase in adjustment costs has little effect on firms' cash demand and R&D choices.

A 5% decrease in  $\rho$  increases innovation volatility and generates a 24% increase in the average cash ratio. It is the most important contributor to high-tech firms' cash demand locally. Knowledge spillover  $\theta$  is the second most important factor in driving high-tech firms' cash policy, followed by innovation efficiency which is captured by a drop in  $\delta$ . A 5% increase in the knowledge spillover externality drives up the mean cash ratio by 20%, and a 5% rise in innovation efficiency induces a 14.5% increase in the average cash ratio.

Surprisingly, equity-issuance costs do not matter much locally. Both the fixed and linear costs ( $\lambda_0$  and  $\lambda_1$ ) play a limited role. Firms on average increase their cash ratios by 1% in response to a 5% increase in linear costs, and are rather insensitive

**Table 5**

**Local Role of R&D Features: Robustness.** This table summarizes the effects of each key R&D feature on firms' R&D and cash choices locally, by changing each parameter in the direction which enhances or strengthens the corresponding mechanism. The parameter moves by the amount which shifts firm value 1% from the benchmark.

|                                                            | Cash  |                          |                          | R&D    |                          |                          | Firm Value |
|------------------------------------------------------------|-------|--------------------------|--------------------------|--------|--------------------------|--------------------------|------------|
|                                                            | Mean  | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | Median | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | % change   |
| <b>Benchmark Model</b>                                     | 0.200 | 0.047                    | 0.257                    | 0.053  | 0.035                    | 0.091                    |            |
| <b>R&amp;D Adjustment Costs</b>                            |       |                          |                          |        |                          |                          |            |
| Fixed adjustment cost ( $\gamma_0 \uparrow 0.072$ )        | 0.198 | 0.047                    | 0.253                    | 0.052  | 0.035                    | 0.088                    | −1%        |
| Convex adjustment cost ( $\gamma_2 \uparrow 2.508$ )       | 0.200 | 0.047                    | 0.253                    | 0.053  | 0.035                    | 0.092                    | −1%        |
| <b>Innovation Uncertainty</b>                              |       |                          |                          |        |                          |                          |            |
| Innovation efficiency ( $\delta \downarrow 0.194$ )        | 0.219 | 0.047                    | 0.251                    | 0.053  | 0.035                    | 0.090                    | 1%         |
| Innovation volatility ( $\rho \downarrow 0.740$ )          | 0.215 | 0.047                    | 0.254                    | 0.053  | 0.035                    | 0.091                    | 1%         |
| <b>Externality</b>                                         |       |                          |                          |        |                          |                          |            |
| Knowledge spillover ( $\theta \uparrow 0.615$ )            | 0.208 | 0.047                    | 0.251                    | 0.052  | 0.035                    | 0.091                    | 1%         |
| <b>Financial Frictions</b>                                 |       |                          |                          |        |                          |                          |            |
| Fixed financing cost ( $\lambda_0 \uparrow 0.050$ )        | 0.200 | 0.047                    | 0.253                    | 0.053  | 0.035                    | 0.093                    | −1%        |
| Linear equity-issuance cost ( $\lambda_1 \uparrow 0.140$ ) | 0.202 | 0.047                    | 0.265                    | 0.053  | 0.035                    | 0.088                    | −1%        |
| <b>Market Competition</b>                                  |       |                          |                          |        |                          |                          |            |
| Elasticity of substitution ( $\sigma \uparrow 2.004$ )     | 0.204 | 0.043                    | 0.284                    | 0.054  | 0.037                    | 0.091                    | −1%        |
| Fixed operating costs ( $c_F \uparrow 0.665$ )             | 0.203 | 0.040                    | 0.273                    | 0.053  | 0.037                    | 0.089                    | −1%        |

to a 5% increase in fixed costs. A small further increase in external financing costs only generates a small increase in the marginal benefit of holding cash and in turn limited rise in precautionary cash demand.

Intensified market competition—a higher elasticity of substitution  $\sigma$  or a higher operating cost  $c_F$ —reduces firms' cash holdings. In particular, a 5% higher elasticity induces a 14.5% drop in average cash ratio, and a 5% rise in fixed operating costs reduces average cash ratio by 12.5%. Both are due to the substantial drop in resources available for cash saving.

However, we should be somewhat cautious in interpreting the elasticities of cash saving to each of the mechanisms above because of the substantial heterogeneity in the changes to firm value. This leads us to the next section on robustness.

#### 4.1.2. Robustness

Although we computed the elasticity of cash saving to parameter changes in the previous subsection, there are alternative ways to compare local effects. One alternative method is to change each parameter such that average firm value moves by exactly 1%. Firm value will go up or down by 1% depending on whether the strengthened channel is beneficial or not to the firms in the economy. Recall that we always enhance each channel for the ease of interpretation. We can then re-evaluate the effects of each factor on firms' cash choices. Results are reported in Table 5. Of course, we could also standardize parameter perturbations by a particular percentage change of other moments such as sales. Average firm value, however, has the nice property that all parameters have an effect on it, which is not true of most other moments.

Overall, our results are quite robust. Higher innovation volatility and innovation efficiency remain the key contributors to high-tech firms' large cash balances. They drive up the average cash ratio by more than 7.5%. Knowledge spillover, on the other hand, increases cash ratios by a smaller amount. Financial frictions and R&D adjustment costs still play a limited role at the margin.

Moreover, the results of market competition in Table 4 are overturned. The effects of market competition on cash holdings are much more muted once we target a 1% firm-value change. And because of non-monotonicity, the rise in market competition increases cash ratio by 1.5–2% in this case.

#### 4.1.3. Global effects

Next, we quantify the effects of each R&D feature globally by shutting them down in turn. We either set the parameter to zero or set the parameter such that the average firm value changes by 25%. We do not push down all parameters to zero because it either doesn't make sense to do so or there is too strong of an effect. For example, setting the innovation efficiency parameter  $\delta$  to zero would actually increase innovation efficiency instead of shutting it down or reducing it. Specifically, we set adjustment costs ( $\gamma_0$  and  $\gamma_2$ ), knowledge spillover ( $\theta$ ), and financial frictions ( $\lambda_0$  and  $\lambda_1$ ) to zero. We raise  $\delta$  and  $\rho$  to reduce the level of innovation efficiency and innovation volatility, respectively, and lower  $\sigma$  and  $c_F$  to reduce the degree of market competition. Each separate parameter movement changes firm value by exactly 25%. Note that reducing innovation efficiency and volatility lowers firm value by 25%, while reducing market competition raises firm value by 25%. Another, perhaps easier, way to understand the global effects is that each parameter moves in the opposite direction in comparison to the parameter movements in the previous sections where the corresponding mechanism was strengthened.

Results are reported in Table 6. To ease the juxtaposition between the importance of each feature in explaining innovative firms' high cash balances, we report the percentage changes in average cash ratio relative to our benchmark model in the third column.

**Table 6**

**Global Role of R&D Features.** This table summarizes the effects of each key R&D feature on firms' R&D and cash choices globally, by shutting down each feature alternately. We set adjustment costs ( $\gamma_0$  and  $\gamma_2$ ), knowledge spillover ( $\theta$ ), and financial frictions ( $\lambda_0$  and  $\lambda_1$ ) to zero. We lower the levels of innovation efficiency ( $\delta$ ), innovation volatility ( $\rho$ ), and market competition ( $\sigma$  and  $c_F$ ) such that each of these parameters changes firm value by 25%. The arrows beside each parameter indicate whether the parameter increases or decreases to shutdown the mechanism. The percentage change in average firm value compared to the benchmark model is reported in the last column.

|                                                                                              | Cash  |          | R&D    |                          |                          | Firm Value |
|----------------------------------------------------------------------------------------------|-------|----------|--------|--------------------------|--------------------------|------------|
|                                                                                              | Mean  | % change | Median | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | % change   |
| <b>Benchmark Model</b>                                                                       | 0.200 |          | 0.053  | 0.035                    | 0.091                    |            |
| <b>R&amp;D Adjustment Costs</b>                                                              |       |          |        |                          |                          |            |
| Fixed adjustment cost ( $\gamma_0 \downarrow 0$ )                                            | 0.243 | 21.5%    | 0.058  | 0.037                    | 0.083                    | 10.5%      |
| Convex adjustment cost ( $\gamma_2 \downarrow 0$ )                                           | 0.239 | 19.5%    | 0.060  | 0.028                    | 0.082                    | 11.4%      |
| Both adjustment costs ( $\gamma_0$ and $\gamma_2 \downarrow 0$ )                             | 0.279 | 39.5%    | 0.056  | 0.028                    | 0.078                    | 10.2%      |
| <b>Innovation Uncertainty</b>                                                                |       |          |        |                          |                          |            |
| Innovation efficiency ( $\delta \uparrow 0.214$ )                                            | 0.182 | −9%      | 0.059  | 0.041                    | 0.097                    | −25%       |
| Innovation volatility ( $\rho \uparrow 1.428$ )                                              | 0.157 | −21.5%   | 0.058  | 0.045                    | 0.087                    | −25%       |
| <b>Externality</b>                                                                           |       |          |        |                          |                          |            |
| Knowledge spillover ( $\theta \downarrow 0$ )                                                | 0.151 | −24.5%   | 0.081  | 0.054                    | 0.140                    | −22.3%     |
| <b>Financial Frictions</b>                                                                   |       |          |        |                          |                          |            |
| Fixed equity-issuance cost ( $\lambda_0 \downarrow 0$ )                                      | 0.166 | −17%     | 0.058  | 0.039                    | 0.104                    | 3.1%       |
| Linear equity-issuance cost ( $\lambda_1 \downarrow 0$ )                                     | 0.168 | −16%     | 0.058  | 0.037                    | 0.102                    | 1.1%       |
| Both equity-issuance costs ( $\lambda_0$ and $\lambda_1 \downarrow 0$ )                      | 0.054 | −73%     | 0.060  | 0.042                    | 0.101                    | 0.04%      |
| Financial frictions and dividend tax ( $\lambda_0$ , $\lambda_1$ and $\tau_d \downarrow 0$ ) | 0     | −100%    | 0.048  | 0.036                    | 0.071                    | 30.9%      |
| <b>Market Competition</b>                                                                    |       |          |        |                          |                          |            |
| Elasticity of substitution ( $\sigma \downarrow 1.952$ )                                     | 0.180 | −10%     | 0.046  | 0.033                    | 0.073                    | 25%        |
| Fixed operating costs ( $c_F \downarrow 0.602$ )                                             | 0.167 | −16.5%   | 0.048  | 0.033                    | 0.073                    | 25%        |

Results in Table 6 suggest that, again, one of the key contributors to high-tech firms' cash stock is innovation volatility associated with the probability of realizing breakthroughs. When we shut down the innovation volatility channel by raising  $\rho$  from 0.76 to 1.43, the chance of moving from median productivity to the technological frontier falls from 4.9% to 1%. Firms then devote more resources to R&D to improve innovation success rates. Higher expenses, together with lower income, leave less resources for cash savings. Firms' average cash ratio drops by 21.5%. Results of these large perturbations also confirm the importance of knowledge spillover and financial frictions. Closing these channels reduces the average cash ratio by 24.5% and 73%, respectively. We also report results for a case without financial frictions and dividend taxes. As shown, firms in this case choose to hold zero cash. This is because the absence of both frictions makes the firm's periodic utility function entirely linear, and firms therefore no longer act as risk-averse agents and demand precautionary cash.

In addition, market competition is another factor that is critical for understanding innovative firms' large cash reserves; yet, it exhibits non-monotonic effects as shown and explained in Subsection 3.2.1.6. Again, the non-monotonicity arises from the interaction between the precautionary motive and the substitution and income effects. When market competition decreases (lower  $\sigma$  or  $c_F$ ), the precautionary motive is dominant and decreases the cash ratio. When market competition increases (higher  $\sigma$  or  $c_F$ ), the substitution and income effects take over. Firms want to raise their R&D expenditures even though they have less cash flow and this draws resources away from cash.

Contrary to the traditional belief, we find that lower R&D adjustment costs lead to a higher cash ratio. The absence of adjustment costs reduces the total expenditures on investment, and in turn, frees up resources which can be converted into cash holdings. Incumbents tend to save out of cash flow even when there is a low probability they will require external finance in the future. In other words, the presence of financial frictions and innovation uncertainty guarantees the value of cash, regardless of the magnitude of R&D adjustment costs. When fixed and/or convex adjustment costs are shut down, cash rises by at least 19.5%.

In sum, Tables 4–6 shed light on the main drivers behind innovative firms' large cash balances. Our results suggest that the main contributors are innovation uncertainty and knowledge spillover, while financial frictions come into play when perturbations are large. Market competition has non-monotonic effects on firms' cash policy, and R&D adjustment costs matter the least. However, each mechanism still has a sizable impact on cash when it is shut down. We must look back at the local effects in the previous section to gauge which factors are the most important at the margin.

#### 4.2. The role of corporate savings

We further answer the question of whether or not internal savings absorb the adverse impacts of financial frictions and by how much. To do so, we perform a counterfactual experiment regarding the role of firms' savings. Firms' cash holdings in our model play a non-trivial role: they not only finance but also compete with R&D investment in allocation.

We conduct this exercise by disallowing firms to save and comparing the resulting R&D distribution with the benchmark case. Since cash holdings have direct impacts on firm value which in turn affects the R&D-to-firm value ratio, we use the

**Table 7**

**The Role of Corporate Savings.** This table shows the role of corporate savings. We solve our benchmark model and the same model without a cash-saving option. We then compare their corresponding model-implied R&D-to-sales distribution, exit rates, average sales, and productivity persistence.

|                               | Benchmark Model | Without Saving |
|-------------------------------|-----------------|----------------|
| <b>R&amp;D-to-sales ratio</b> |                 |                |
| Median                        | 0.140           | 0.122          |
| 1 <sup>st</sup> quartile      | 0.068           | 0.078          |
| 3 <sup>rd</sup> quartile      | 0.186           | 0.160          |
| Mean                          | 0.155           | 0.129          |
| <b>Market competition</b>     |                 |                |
| Exit rate                     | 0.019           | 0.028          |
| <b>Sales</b>                  |                 |                |
| Mean                          | 2.834           | 2.824          |
| <b>Productivity</b>           |                 |                |
| Persistence                   | 0.824           | 0.845          |

R&D-to-sales ratio instead to avoid the mechanical change in the R&D-to-firm value distribution. Results are reported in Table 7.

As shown, without the option of saving cash to transfer resources across time, the stationary distribution of R&D changes. While the 1<sup>st</sup> quartile of R&D-to-sales ratio increases by 14.7%, the median, 3<sup>rd</sup> quartile, and the average decline by 12.9%, 14.0%, and 16.8%, respectively. The former stems from the rise in market exit rates, which induces the entry of firms with relatively high productivity and high R&D spending. The latter indicates that savings, on average, do help to correct the distortions created by financial frictions, especially for R&D intensive incumbents, and the magnitude is large.

Moll (2014) explores the role of corporate savings in undoing capital misallocation. He demonstrates that TFP losses decrease in shock persistence and the effect is very steep. That is, corporate savings play a limited role in mitigating capital misallocation if productivity shocks are not sufficiently persistent, because firms cannot accumulate enough wealth to self finance. Buera et al. (2011) reach a similar conclusion. Midrigan and Xu (2010) also find substantial productivity losses in the intensive margin when they remove permanent shocks, and instead, introduce transitory shocks with high persistence.

Our model considers endogenous productivity evolution, with the model-implied productivity autocorrelation to be 0.824 which falls within the range of a medium level of persistence. The estimated results suggest that even in the case without sufficiently persistent productivity, the model financing scheme is able to reduce sizable distortions from financial frictions and pushes up the average R&D-to-sales ratio from 12.9% to 15.5%. We derive a result different from previous studies, because in our model firms are allowed to issue equity each period and can accumulate internal funds by keeping the proceeds from external finance, and this feature is often absent in previous studies. Of course, the cash holdings in the high-tech sector fail to completely offset the negative impacts of financial frictions.

We should emphasize that providing external finance is still rational from the investors' point of view. Investors know that if a firm is committed to investing in R&D, its future prospects are quite good even if the current productivity level is relatively low. The firm would incur a large deadweight loss if it is forced to reduce R&D investment unnecessarily. This behavior of financing unproductive firms, is evidenced by the fact that, in our Compustat sample, over half of high-tech firms have negative net income. For high-tech firms with less than 1 billion 2010 dollars in total assets, the average net income is even negative.

Furthermore, disallowing firms to save also reduces average sales and drives up productivity persistence. The former arises from the drop in real outputs which cannot be offset by the rise in product price following the large fall in R&D investment. The latter is generated from the fact that R&D becomes the only instrument to transfer resources over time. When firms devote all available resources to R&D, R&D becomes more persistent which in turn increases productivity persistence.

#### 4.3. Model re-estimation with productivity moments

Finally, we examine the robustness of our benchmark parameter estimates and the corresponding model performance in matching productivity targets. We remove the first and third quartile of the R&D ratio as well as the serial correlation of the R&D ratio from our benchmark estimation, but keep the productivity moments. The mean, standard deviation, and autocorrelation of productivity serve as good proxies for the removed R&D moments and the model becomes just-identified. The parameter identification arguments previously discussed would also transfer nicely to the productivity moments. New parameter estimates and targeted moments are reported in Table 8.

Overall, the new set of parameter estimates is close to our benchmark, and targeted productivity and cash moments are better matched. Specifically, we find that a few key parameter estimates slightly deviate from those in our benchmark estimation to better explain cash and productivity dynamics. First, financial frictions become stronger, implied by the larger

**Table 8**

**Model Re-estimation: Parameters and Targeted Moments.** This table summarizes the estimation results by jointly matching nine selected data moments. Panel A presents the parameter estimates. Standard errors are presented in parentheses. Panel B reports the nine targeted moments used for model re-estimation. Data moments are calculated based on a sample of Compustat high-tech firms over the period 2000–2014. The constructed weighting matrix minimizes mean squared percent deviations between data and model moments.

| Panel A: Parameters Estimated by SMM                       |        |          |
|------------------------------------------------------------|--------|----------|
| Fixed costs of external finance ( $\lambda_0$ )            | 0.0586 | (0.0110) |
| Linear costs of external finance ( $\lambda_1$ )           | 0.1392 | (0.0400) |
| Fixed adjustment cost ( $\gamma_0$ )                       | 0.0454 | (0.0012) |
| Convex adjustment cost ( $\gamma_2$ )                      | 2.4130 | (0.0749) |
| Innovation efficiency ( $\delta$ )                         | 0.2502 | (0.0003) |
| Knowledge spillover ( $\theta$ )                           | 0.7116 | (0.0004) |
| Innovation volatility ( $\rho$ )                           | 1.0200 | (0.0009) |
| Fixed operating cost ( $c_F$ )                             | 0.5952 | (0.0002) |
| Wage cost ( $w$ )                                          | 0.2085 | (0.0044) |
| Panel B: Simulated Model Moments: Targeted Moments         |        |          |
|                                                            | Data   | Model    |
| (i) R&D:                                                   |        |          |
| Fraction of firms with zero R&D investment                 | 0.021  | 0.044    |
| (ii) Cash:                                                 |        |          |
| 25 <sup>th</sup> percentile of cash-to-firm value ratio    | 0.061  | 0.057    |
| 75 <sup>th</sup> percentile of cash-to-firm value ratio    | 0.249  | 0.254    |
| Serial correlation of within-firm cash-to-firm value ratio | 0.693  | 0.649    |
| Correlation with R&D-to-firm value ratio                   | 0.369  | 0.389    |
| (iii) Productivity:                                        |        |          |
| Industry average productivity                              | 1.000  | 1.105    |
| Cross-sectional dispersion of productivity                 | 0.561  | 0.639    |
| Serial correlation of within-firm productivity             | 0.797  | 0.826    |
| (iv) Market competition:                                   |        |          |
| Exit rate                                                  | 0.019  | 0.029    |

estimates of fixed and linear external financing costs. These changes produce a closer match for the autocorrelation of cash. The estimate of  $\delta$  also becomes larger. This suggests that innovation becomes systematically less efficient, which leads to a lower average productivity level and a less dispersed productivity distribution. The increase in knowledge spillover  $\theta$  contributes to the improved match of the 25<sup>th</sup> percentile of the cash-ratio distribution, but it also undesirably generates a larger share of zero R&D performers because more industry followers choose to free-ride. This leads to smaller estimates for R&D adjustment costs. The innovation volatility  $\rho$  also increases. Its rise implies a higher likelihood to make incremental innovations and generates a bit more persistent within-firm productivity.

To further evaluate our model performance in reproducing firms' R&D behavior, we report nontargeted R&D moments together with cash moments in Table 9. As shown, the R&D distribution remains well characterized, although the distance between data and model-implied 25<sup>th</sup> and 75<sup>th</sup> percentile of R&D widens. The latter is possibly driven by the larger estimate of  $\rho$ . When the likelihood of reaching extremely high or low productivity levels drops substantially, realized productivity becomes more persistent. As a result, leaders' marginal benefit of R&D investment falls and they invest less. This effectively increases followers' probability of catching up with leaders and prompts followers to invest more in R&D. In addition, both the top and bottom 1<sup>st</sup> and 5<sup>th</sup> percentiles of the R&D distribution match data better.

The empirical cash distribution is also well replicated by our model. Compared to the benchmark, both cross-sectional dispersion and industry median cash ratios are closer to their empirical counterparts, although at the expense of a bit higher but still well-matched industry average cash ratio.

Finally, we perform the same exercise as Table 4 and show the local effects of each R&D feature with the new set of parameter estimates. As evidenced by Table 10, the local importance of each model feature in explaining high-tech firms' cash holdings is very similar to that found in our benchmark case. Innovation uncertainty and knowledge spillover remain the major factors. We again demonstrate the robustness of our core results.

## 5. Conclusion

High-tech firms hold a considerable amount of cash. A better understanding of their joint R&D and cash-saving choices can help design policies to promote R&D investment and innovation. In this paper, we systematically analyze how some unique features of R&D shape R&D and cash decisions and assess the importance of corporate saving in tempering the distortions from financial frictions.



**Table 9**

**Simulated Model Moments: Nontargeted Moments.** This table presents nontargeted moments for the R&D-to-firm value ratio and cash-to-firm value ratio. Data moments are calculated based on a sample of Compustat high-tech firms over the period 2000–2014.

| Moments                                                   | Data  | Model |
|-----------------------------------------------------------|-------|-------|
| (i) R&D:                                                  |       |       |
| Industry average R&D-to-firm value ratio                  | 0.081 | 0.090 |
| Cross-sectional dispersion of R&D-to-firm value ratio     | 0.109 | 0.126 |
| Industry median R&D-to-firm value ratio                   | 0.048 | 0.061 |
| Serial correlation of within-firm R&D-to-firm value ratio | 0.518 | 0.524 |
| 1-percentile of R&D-to-firm value ratio                   | 0.000 | 0.000 |
| 5-percentile of R&D-to-firm value ratio                   | 0.003 | 0.012 |
| 10-percentile of R&D-to-firm value ratio                  | 0.008 | 0.034 |
| 25-percentile of R&D-to-firm value ratio                  | 0.021 | 0.044 |
| 75-percentile of R&D-to-firm value ratio                  | 0.100 | 0.085 |
| 90-percentile of R&D-to-firm value ratio                  | 0.181 | 0.152 |
| 95-percentile of R&D-to-firm value ratio                  | 0.259 | 0.251 |
| 99-percentile of R&D-to-firm value ratio                  | 0.517 | 0.559 |
| (ii) Cash:                                                |       |       |
| Industry average cash-to-firm value ratio                 | 0.193 | 0.204 |
| Cross-sectional dispersion of cash-to-firm value ratio    | 0.211 | 0.204 |
| Industry median cash-to-firm value ratio                  | 0.135 | 0.185 |
| 1-percentile of cash-to-firm value ratio                  | 0.000 | 0.000 |
| 5-percentile of cash-to-firm value ratio                  | 0.008 | 0.000 |
| 10-percentile of cash-to-firm value ratio                 | 0.021 | 0.000 |
| 90-percentile of cash-to-firm value ratio                 | 0.424 | 0.422 |
| 95-percentile of cash-to-firm value ratio                 | 0.580 | 0.547 |
| 99-percentile of cash-to-firm value ratio                 | 1.006 | 1.023 |

**Table 10**

**Local Role of R&D Features.** This table summarizes the effects of each key R&D feature on firms' R&D and cash choices locally, by changing the value of each parameter in Table 8 by 5% in the direction which enhances or strengthens the mechanism. The arrows beside each parameter indicate whether the parameter increases or decreases to strengthen the mechanism. The percentage change in average firm value compared to the benchmark model is reported in the last column.

|                                                             | Cash  |                          |                          | R&D    |                          |                          | Firm Value |
|-------------------------------------------------------------|-------|--------------------------|--------------------------|--------|--------------------------|--------------------------|------------|
|                                                             | Mean  | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | Median | 1 <sup>st</sup> quartile | 3 <sup>rd</sup> quartile | % change   |
| <b>Benchmark Model</b>                                      | 0.204 | 0.057                    | 0.254                    | 0.061  | 0.044                    | 0.085                    |            |
| <b>R&amp;D Adjustment Costs</b>                             |       |                          |                          |        |                          |                          |            |
| Fixed adjustment cost ( $\gamma_0 \uparrow 0.0476$ )        | 0.201 | 0.048                    | 0.254                    | 0.063  | 0.044                    | 0.090                    | 0.83%      |
| Convex adjustment cost ( $\gamma_2 \uparrow 2.534$ )        | 0.202 | 0.048                    | 0.254                    | 0.062  | 0.044                    | 0.089                    | 0.14%      |
| <b>Innovation Uncertainty</b>                               |       |                          |                          |        |                          |                          |            |
| Innovation efficiency ( $\delta \downarrow 0.238$ )         | 0.221 | 0.098                    | 0.282                    | 0.058  | 0.043                    | 0.090                    | 20.7%      |
| Innovation volatility ( $\rho \downarrow 0.969$ )           | 0.222 | 0.056                    | 0.288                    | 0.060  | 0.043                    | 0.088                    | 4.5%       |
| <b>Externality</b>                                          |       |                          |                          |        |                          |                          |            |
| Knowledge spillover ( $\theta \uparrow 0.747$ )             | 0.217 | 0.056                    | 0.288                    | 0.060  | 0.043                    | 0.082                    | 2.2%       |
| <b>Financial Frictions</b>                                  |       |                          |                          |        |                          |                          |            |
| Fixed equity-issuance costs ( $\lambda_0 \uparrow 0.0615$ ) | 0.205 | 0.057                    | 0.254                    | 0.061  | 0.044                    | 0.085                    | −0.09%     |
| Linear equity-issuance cost ( $\lambda_1 \uparrow 0.146$ )  | 0.204 | 0.057                    | 0.254                    | 0.061  | 0.044                    | 0.085                    | −0.05%     |
| <b>Market Competition</b>                                   |       |                          |                          |        |                          |                          |            |
| Elasticity of substitution ( $\sigma \uparrow 2.100$ )      | 0.169 | 0.000                    | 0.142                    | 0.074  | 0.047                    | 0.113                    | −36.7%     |
| Fixed operating costs ( $c_F \uparrow 0.693$ )              | 0.140 | 0.019                    | 0.184                    | 0.069  | 0.051                    | 0.100                    | −20.4%     |

The model embeds five key factors highly pertinent for high-tech industries: R&D adjustment costs, financial frictions, knowledge spillover, innovation uncertainty, and market competition. We show that our estimated model successfully replicates the empirical patterns of industry-level and firm-level R&D investment and cash-saving choices. We then use the validated model to determine the quantitative importance of relevant R&D features in explaining high-tech firms' substantial cash reserves. We find the following. Innovation uncertainty matters the most for the observed strong saving demand, followed by knowledge spillover and financial frictions. R&D adjustment costs have a relatively weak relationship with cash policy, and market competition has non-monotonic effects on cash. We also find that corporate saving plays an important role in correcting distortions from financial frictions, even though cash holdings cannot fully offset the negative impacts.

While our paper identifies and evaluates the main drivers of U.S. high-tech firms' large cash stocks, the analyses can be improved and extended. One direction is to consider a general equilibrium model that incorporates the saving behavior

of the corporate sector with a household problem and/or investigate the cash, R&D, and innovation relationship in other countries such as Gao and Zhao (2020) does for China. These extensions, although challenging, would be interesting in light of the empirical evidence in Chen et al. (2017) that the dramatic rise in corporate savings coincides with a substantial fall in household savings in a large majority of countries. Some other challenging extensions worth pursuing are to introduce, derive, and estimate endogenous relationships between various factors.

## Appendix A

### A.1. Model solution

First recall that firm  $i$ 's output  $y_i$  is a function of productivity shock  $z_i$  and labor  $l_i$ :

$$y_i = z_i l_i.$$

A relationship between aggregate price  $P$  and individual price  $p_i$  is derived by minimizing the cost of acquiring a fixed amount of aggregate good  $\bar{Y}$ . Therefore, we want to solve the following minimization problem:

$$\min_{y_i} \{PY\} = \min_{y_i} \left\{ \sum_i p_i y_i \right\}$$

subject to,

$$\left( \sum_i y_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} = \bar{Y}.$$

The Lagrangian is expressed as,

$$\mathcal{L} = \sum_i p_i y_i + \eta \left[ \bar{Y} - \left( \sum_i y_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \right]$$

where  $\eta$  is the Lagrange multiplier. The first-order condition with respect to  $y_i$  is,

$$p_i = \eta \left( \sum_i y_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{1}{\sigma-1}} y_i^{-\frac{1}{\sigma}},$$

which then simplifies to,

$$p_i^\sigma y_i = P^\sigma Y,$$

since the Lagrange multiplier  $\eta$  is equal to  $P$ . Then to find the profit maximizing price, we solve

$$\max_{p_i} \{\pi_i\} = \max_{p_i} \{p_i y_i - w l_i\}.$$

The first order condition with respect to  $p_i$  gives the optimal price as a markup over marginal cost,

$$p_i = \frac{\sigma}{\sigma-1} \frac{w}{z_i}.$$

Finally, profit can be rewritten as,

$$\begin{aligned} \pi_i &= p_i y_i - w l_i - c_f + r c_i - d_i \\ &= (P^\sigma Y) p_i^{1-\sigma} - (P^\sigma Y) w p_i^{-\sigma} z_i^{-1} - c_f + r c_i - d_i \\ &= (P^\sigma Y) \sigma^{-\sigma} (\sigma-1)^{\sigma-1} w^{1-\sigma} z_i^{\sigma-1} - c_f + r c_i - d_i \\ &= (P^\sigma Y) \left( \frac{(\sigma-1) z_i}{\sigma w} \right)^{\sigma-1} \frac{1}{\sigma} - c_f + r c_i - d_i. \end{aligned}$$

Without loss of generality and for simplicity, we normalize  $P^\sigma Y$  to 1 rather than  $w$  to 1 as in Melitz (2003).

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